Business Cycle Accounting

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Business Cycle Accounting

- Preliminary data analysis technique

- Goals:

  - Isolate promising classes of models/theories/stories

  - Guide development of theory

  - Avoid critiques of structural VARs

Idea of Approach

- Equivalence results:
  - Detailed models with frictions observationally equivalent to
  - Growth model with time-varying TFP and tax rates ("wedges")
- Accounting procedure:
  - Use theory plus data to measure wedges
  - Estimate stochastic process governing expectations
  - Feed wedges back one at a time and in combinations
  - How much of output, investment, labor accounted for by each?

Outline

- Business cycle accounting approach
  - Describe prototype growth model
  - Equivalence results
  - Application to Great Depression
- Should replace popular alternative (SVARs)
  - Describe structural VAR procedure
  - Critique procedure using growth model
  - Application to post-war business cycles

Prototype Growth Model

Consumption (c), labor (l), investment (x) solve

$$\max_{\{c_t, l_t, x_t\}} E \sum_{t=0}^{\infty} \beta^t U(c_t, l_t)$$

subject to

$$c_t + (1 + \tau_{xt}) x_t \leq (1 - \tau_{lt}) w_t l_t + r_t k_t + T_t$$

$$k_{t+1} = (1 - \delta) k_t + x_t$$

$$y_t = A_t F(k_t, \gamma^t l_t)$$

Equations for Prototype Growth Model

- Efficiency wedge:

$$y_t = A_t F(k_t, \gamma^t l_t)$$

- Labor wedge:

$$-\frac{U_{lt}}{U_{ct}} = (1 - \tau_{lt})(1 - \alpha) y_t / l_t$$

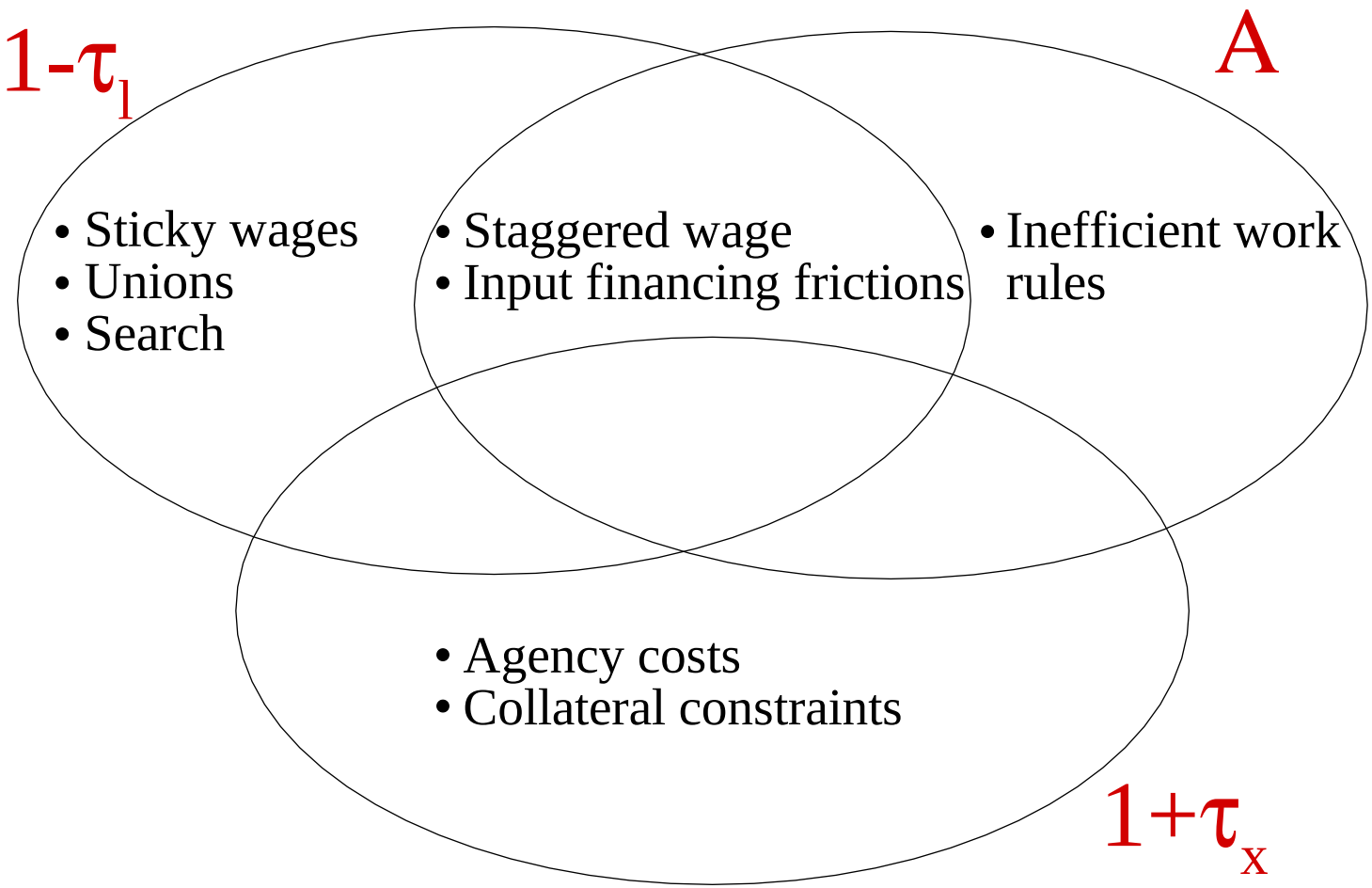
- Investment wedge:

$$(1 + \tau_{xt}) U_{ct} = \beta E_t U_{ct+1} \left[\alpha y_{t+1} / k_{t+1} + (1 + \tau_{xt+1})(1 - \delta) \right]$$

Business Cycle Accounting

- Estimate stochastic processes for A, τ_l, τ_x
- Compute equilibria for prototype economy
- Generate realization by feeding in
  - A only
  - $1 - \tau_l$ only
  - $1 + \tau_x$ only
  - combinations

Mapping Between Original and Prototype Models



Equivalence Results

Efficiency Wedges

$$y_t = A_t F(k_t, \gamma^t l_t)$$

- Changes in blueprints
- Misallocation of inputs across tasks/production units
  - within firms (work rules)
  - across firms (input financing frictions)

Example: Input Financing Frictions

- Final goods producers have technology $y = y_1^{1/2} y_2^{1/2}$

$$\max_{y_1, y_2} \; y - p_1 y_1 - p_2 y_2$$

- Intermediate goods producers have technology $y_i = l_i^{\alpha}$

$$\max_{l_i} \; p_i y_i - R(1 + \tau_i) w l_i$$

- Households supply labor $l = l_1 + l_2$ and solve

$$\max_{c,l} \; U(c, l) \text{ subject to } c = wl + \pi$$

Equivalent Prototype Economy

- Firm solves $\max_l \; Al^\alpha - wl$

- Household solves $\max_{c,l} \; U(c,l)$ subject to $c = (1-\tau)wl + \pi$

- Proposition: Allocations in two economies same if

$$A = \frac{(1+\tau_1)^{\frac{\alpha}{2}}(1+\tau_2)^{\frac{\alpha}{2}}}{[(1+\tau_1)+(1+\tau_2)]^\alpha}$$

$$1-\tau = \frac{1}{2}\left(\frac{1}{1+\tau_1}\right) + \frac{1}{2}\left(\frac{1}{1+\tau_2}\right)$$

Note: Increases in financing distortions makes TFP fall

Labor wedges

$$\frac{U_{lt}}{U_{ct}} = (1 - \tau_{lt}) F_{lt}$$

- Sticky wages
  - Labor market wedges in prototype model
  - Staggering yields efficiency wedges
- Cartels/unions
  - Labor market wedges in prototype model

Example: Sticky Wages

- Economy:
  - Event s_t, $s^t = (s_0, \ldots, s_t)$
  - Stochastic money growth $\mu(s^t)$
  - Utility $U(c, l, m) = u(c, l) + v(m)$
  - Production $F(k, l)$

- Define $\tau_l^*(s^t)$

$$\tau_l^*(s^t) = 1 - \frac{U_l^*(s^t)}{U_c^*(s^t)} \frac{1}{F_l^*(s^t)}$$

where '$*$' indicates equilibrium values

Equivalent Prototype Economy

- Real prototype economy with
  - Stochastic labor taxes $\tau_l(s^t)$
  - Utility $u(c, l)$
  - Production $F(k, l)$
- Proposition. Allocations in the two economies same if $\tau_l(s^t) = \tau_l^*(s^t)$.

Investment wedges

$$(1+\tau_{xt})U_{ct} = \beta E_t U_{ct+1}\left[\alpha y_{t+1}/k_{t+1} + (1+\tau_{xt+1})(1-\delta)\right]$$

- Models with financial frictions, e.g.,
  - Bernanke-Gertler
  - Carlstrom-Fuerst
  - Kiyotaki-Moore
- Map into prototype with investment wedges (See our appendix)

Application to the Great Depression

Measuring Wedges

- Stochastic Process for wedges $s_t = [\log A_t, \tau_{lt}, \tau_{xt}]$

$$s_{t+1} = P_0 + P s_t + Q \eta_{t+1}$$

- Preferences and technology
  - $U(c, l) = \log c + \psi \log(1 - l)$
  - $F(k, l) = A k^{\theta} l^{1-\theta}$

- With data for 1901-1940
  - fix parameters of technology and preferences
  - compute MLE estimates of P_0, P, Q

Recovering Wedges

- Model decision rules are $c(s_t, k_t), x(s_t, k_t), l(s_t, k_t)$
- Set
  - $c(s_t, k_t) = c_t^{DATA}$
  - $x(s_t, k_t) = x_t^{DATA}$
  - $l(s_t, k_t) = l_t^{DATA}$

  with k_t defined recursively from accumulation equation
- Solve for values of $s_t = [\log A_t, \tau_{lt}, \tau_{xt}]$
- Inputting these values gives exactly same series as in data

Figures 1–4
Examining the U.S. Great Depression
Annually, 1929–39; Normalized to Equal 100 in 1929

Figure 1
U.S. Output and Three Measured Wedges

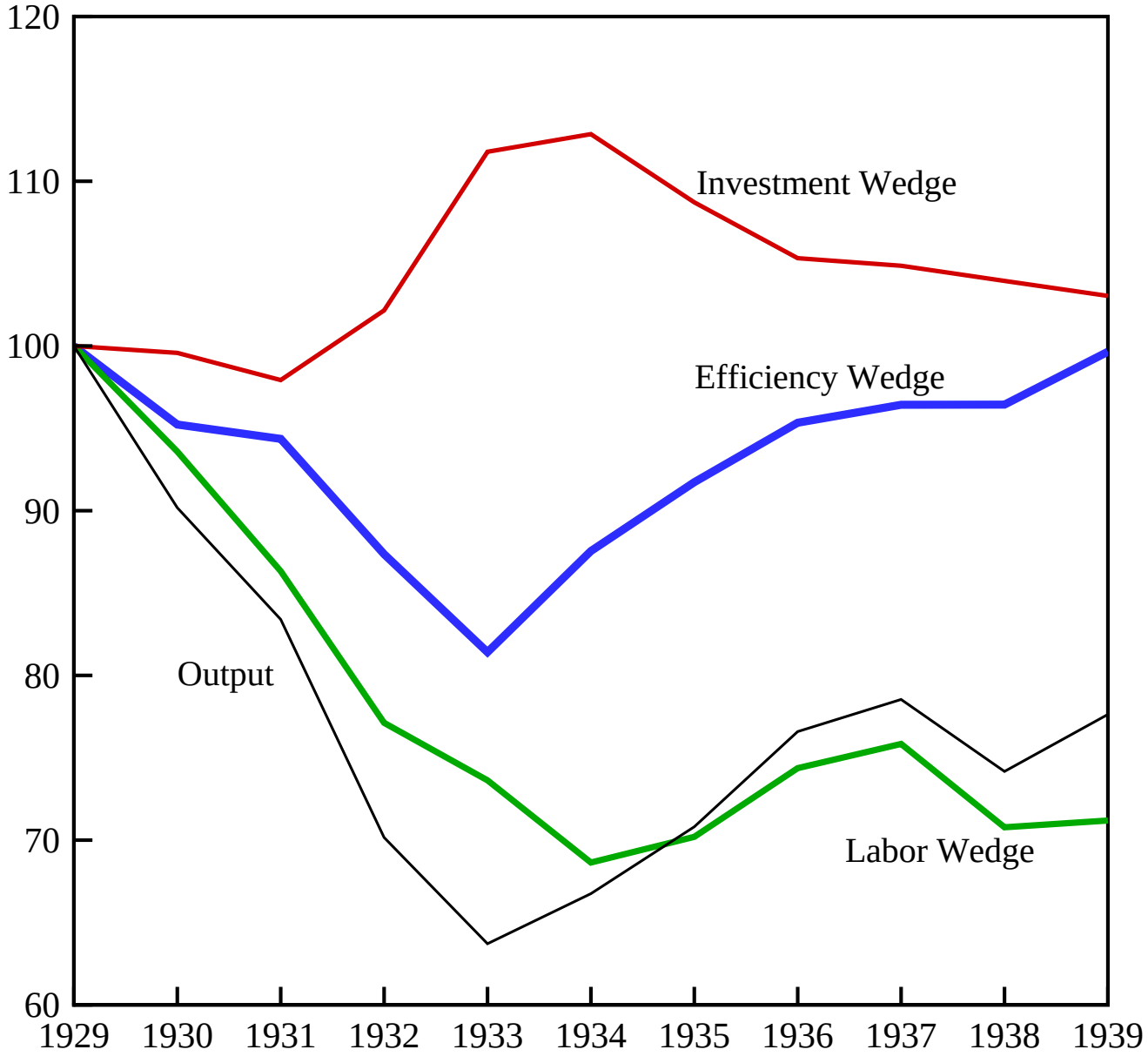


Figure 2
Data and Predictions of Models With Just One Wedge

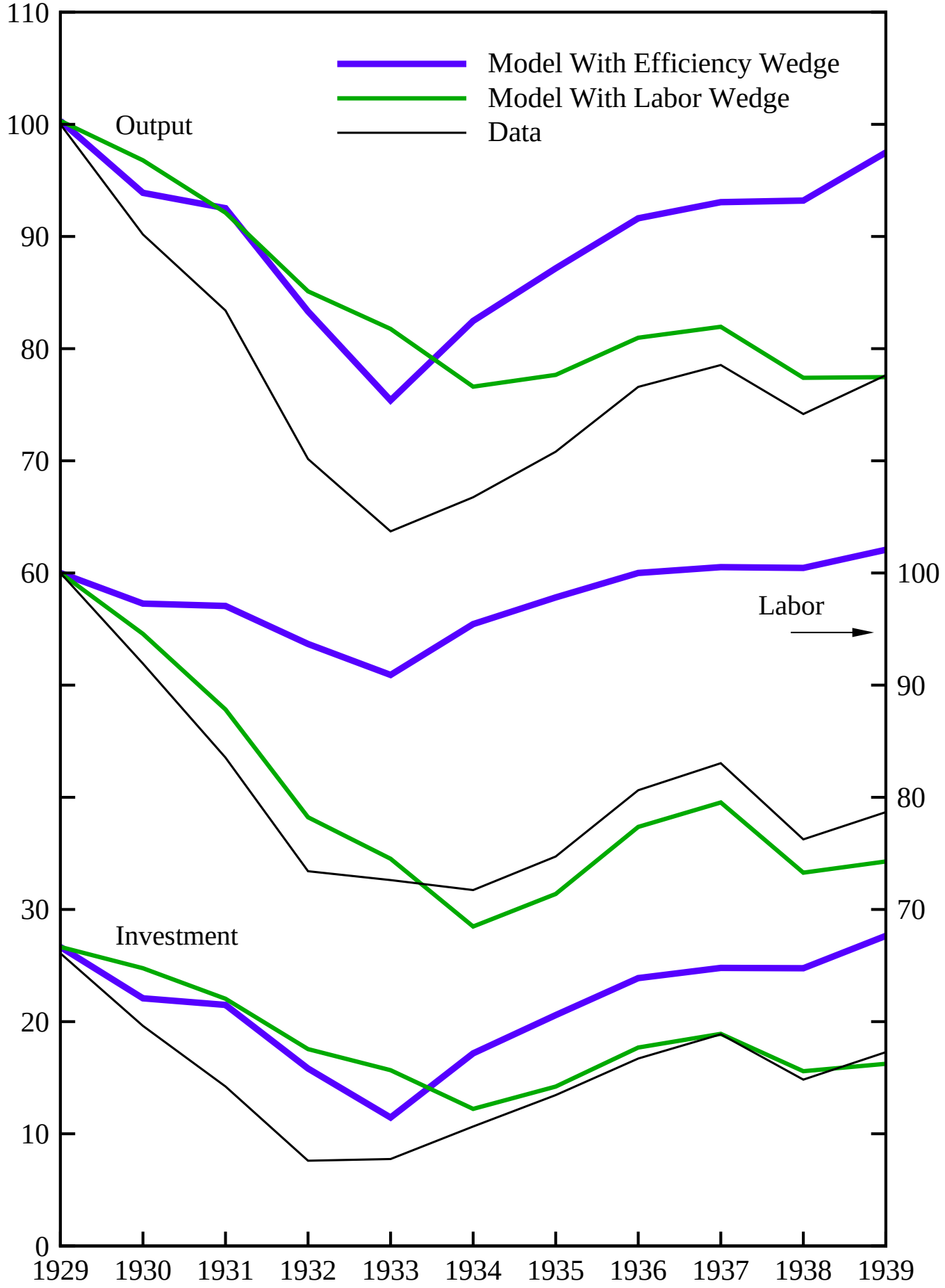


Figure 3
Data and Predictions of Model With Just the Investment Wedge

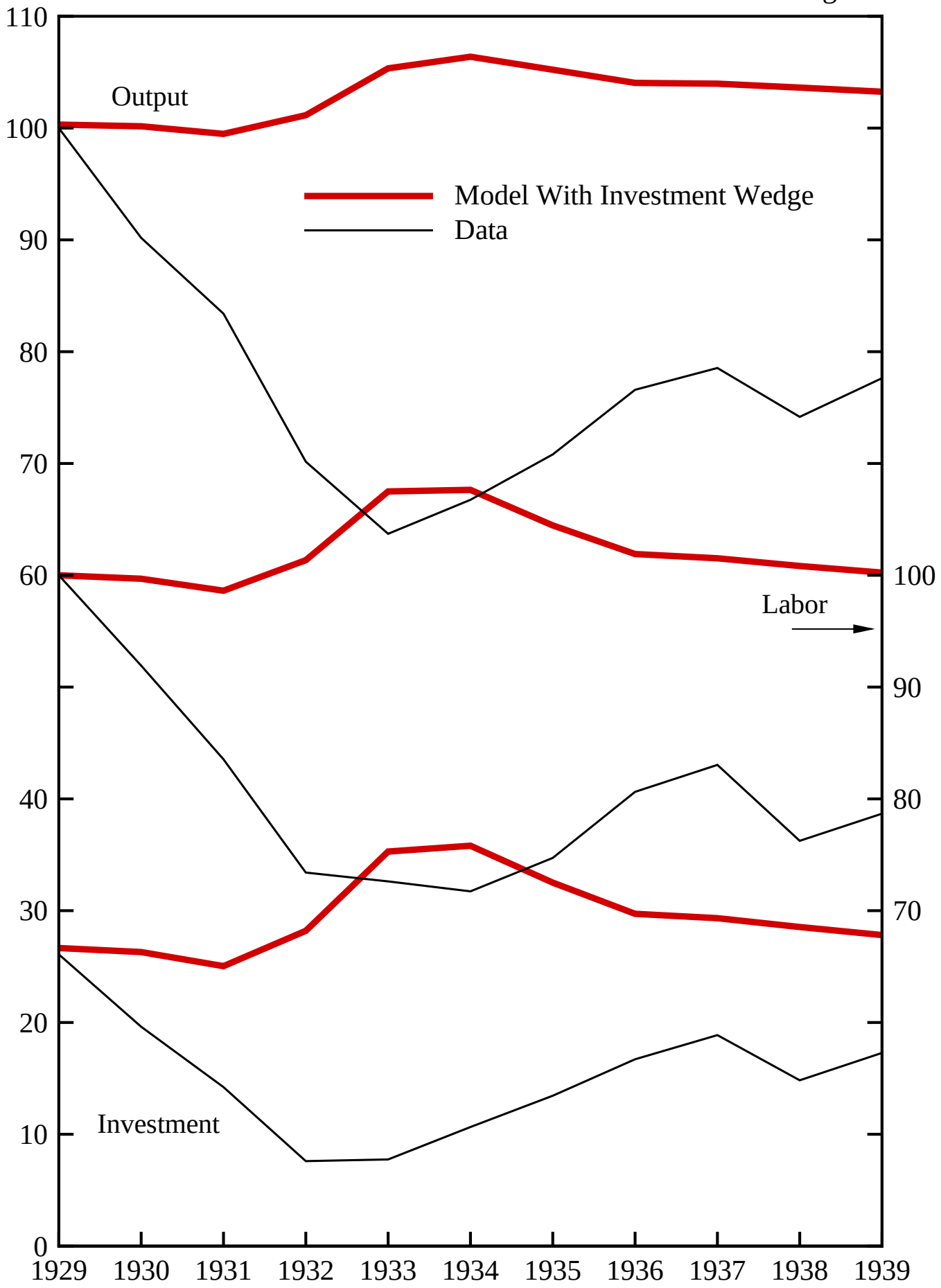
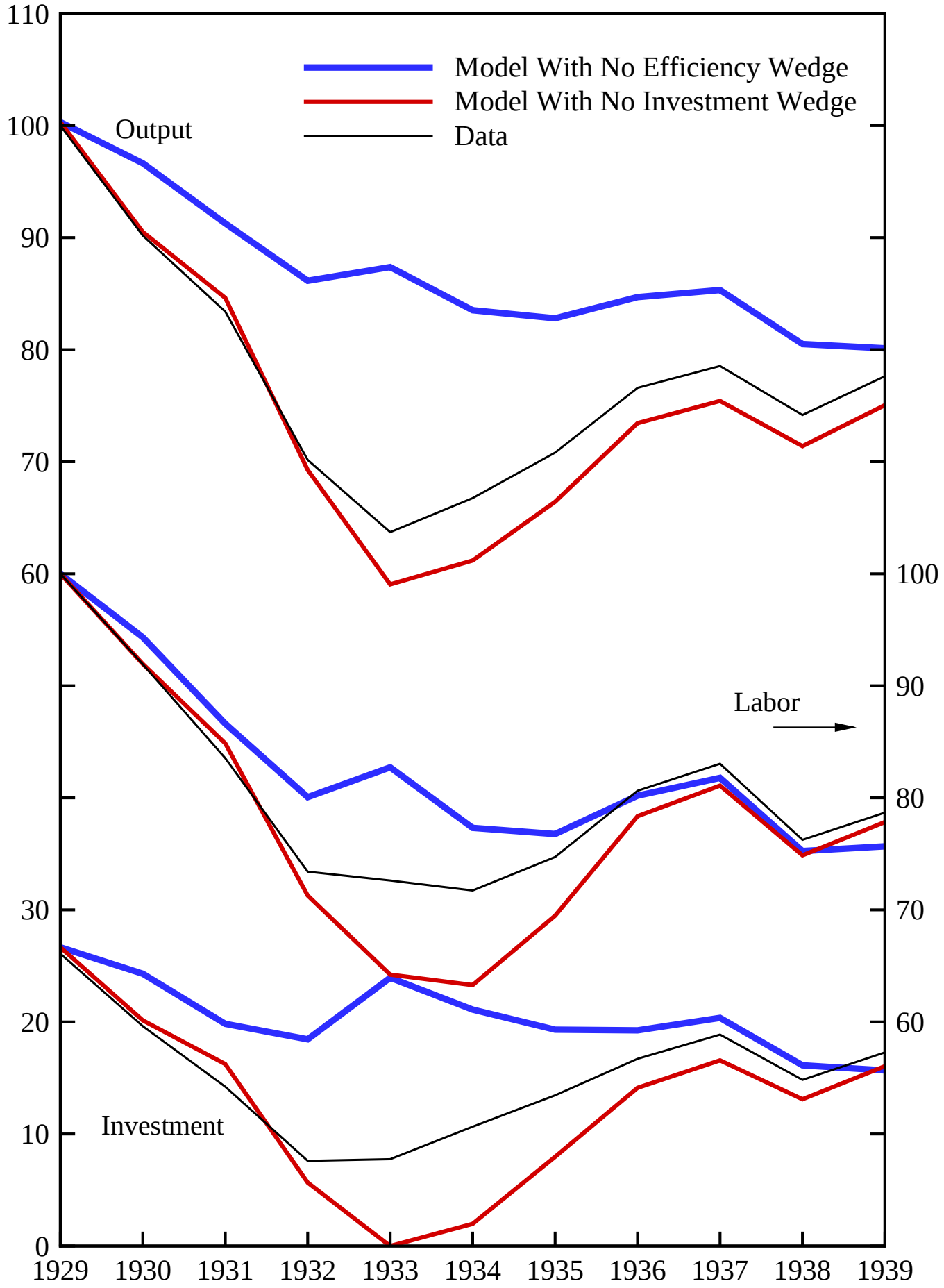


Figure 4
Data and Predictions of Models With All But One Wedge



Sensitivity to the Labor Elasticity

- Index the economy by i:

$$U(c,l) = \frac{c^{1-\sigma}}{1-\sigma} - \frac{l^{1+\zeta_i}}{1+\zeta_i}$$

$$\frac{U_l}{U_c} = (1-\tau_i)F_l \iff l^{\zeta_i} c^{\sigma}(1-\tau_i)(1-\theta)y/l$$

- Proposition: the ζ_2-economy has same equilibrium as the ζ_1-economy

  Proof: $1-\tau_2 = (1-\tau_1)l^{\zeta_2-\zeta_1}$

- Punchline: Different measured wedges—but same equilibrium

Sensitivity to the Capital Utilization

- Define technology as

$$y = A(kh)^{\theta}(nh)^{1-\theta}$$

where n=employed, h=hours, $l = nh$

- Fixed capital utilization: h constant, n varies

$$y = Ak^{\theta}l^{1-\theta}$$

- Variable capital utilization: h varies, n constant

$$y = Ak^{\theta}l^{1-\theta}$$

Sensitivity to the Capital Utilization

- Index the economy by i:

$$y = A_i k^{\theta} l^{\gamma_i} \quad \gamma_i = 1 - \theta \text{ or } 1$$

$$\frac{U_l}{U_c} = (1 - \tau_i)\gamma_i \frac{y}{l}$$

- Proposition: the γ_2-economy has same equilibrium as the γ_1-economy

  Proof: $A_2 = A_1 l^{\gamma_1 - \gamma_2}$ and $1 - \tau_2 = (1 - \tau_1)\gamma_1/\gamma_2$

- Punchline: Different measured wedges—but same equilibrium

Figures 9–12
Varying the Capital Utilization Specification
Great Depression Period, 1929–39

Figure 9
Measured Efficiency Wedges for Two Capital Utilization Specifications

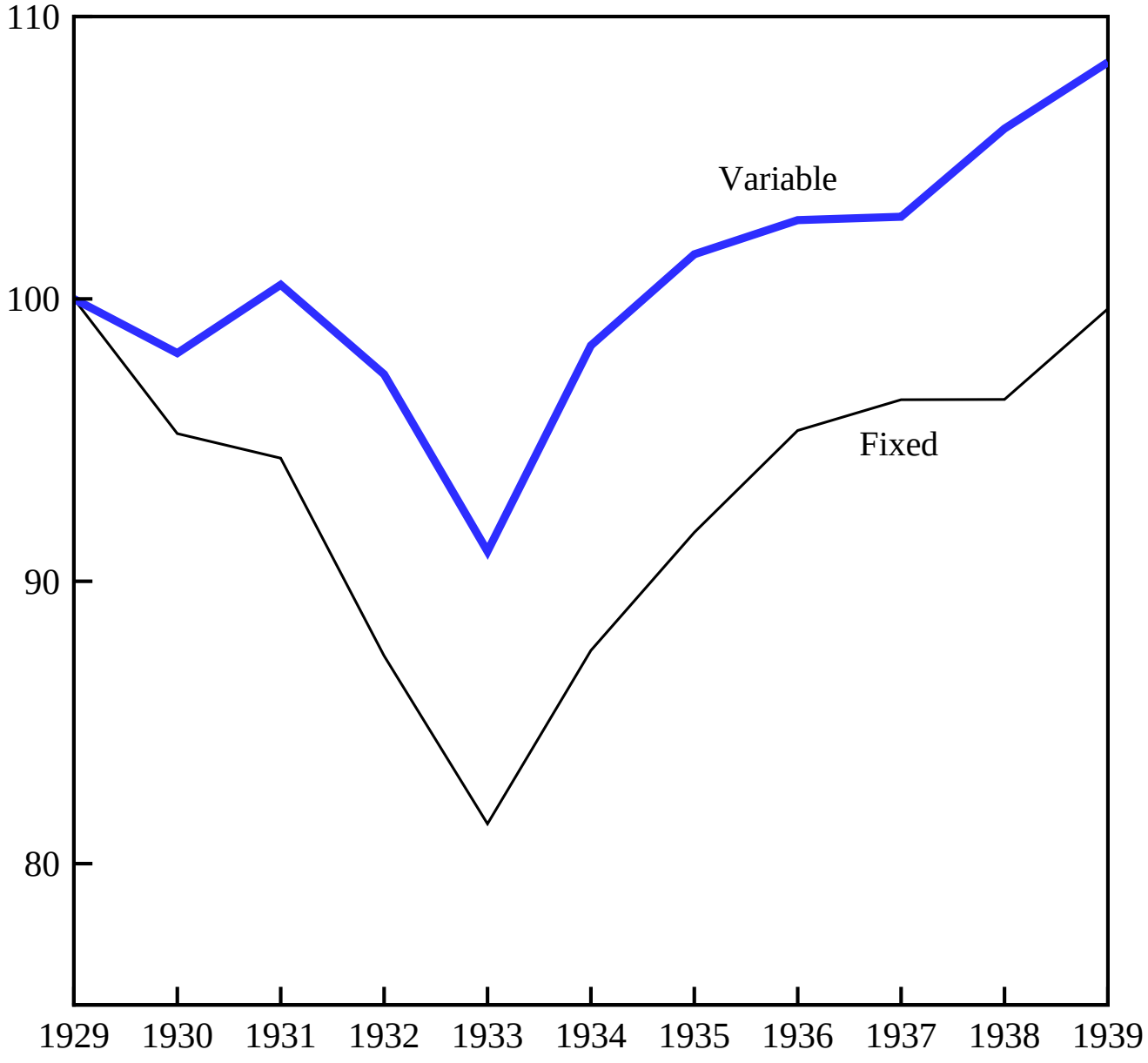
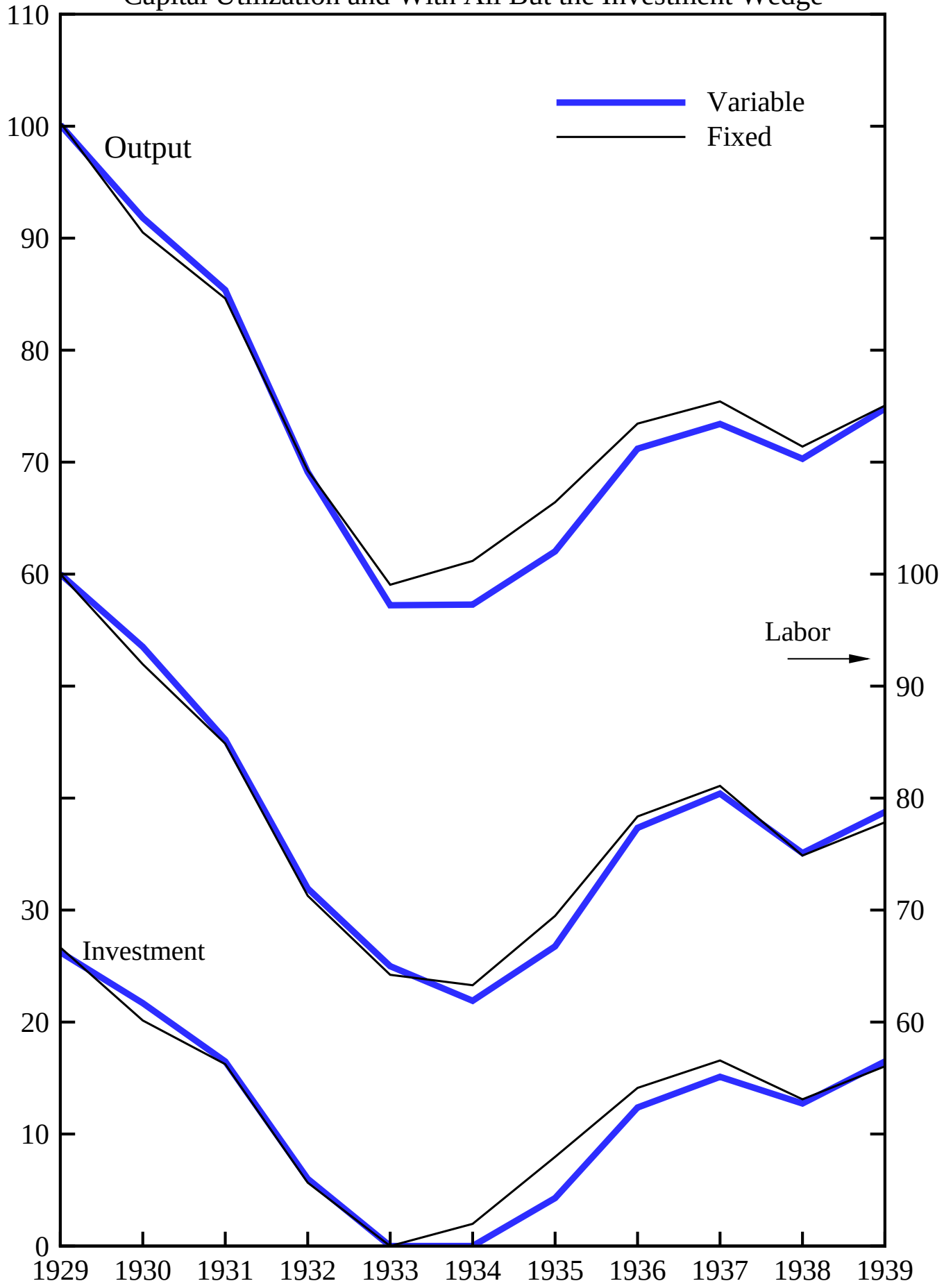


Figure 12
Predictions of Models with Fixed and Variable
Capital Utilization and With All But the Investment Wedge



Recap

- Approach useful for isolating relevant class of theories

- Our application to Great Depression shows
  - Labor and efficiency wedges generate almost all variation
  - While investment wedges are the focus of the literature
  - Size of wedges not relevant—equilibrium responses are.

Popular Alternative: Structural VARs

Current Practice: Structural VARs

- Provide summaries of facts to guide theorists, e.g.,
  - what happens after a technology shock?
  - what happens after a monetary shock?
- Impulse responses used to identify promising classes of models, e.g.,
  - If SVAR finds positive technology shock leads to fall in hours
  - Points to sticky price models (not RBC models) as promising
- SVARs are used a lot . . . but are they useful guides for theory?

Our Evaluation of SVARs Using Growth Model

- Use standard growth model
- Plot **theoretical** impulse response from model
- Generate data from model and apply SVAR procedure
- Plot **empirical** impulse response identified by SVAR procedure
- Compare responses

Main Findings

- Using growth model with SVAR assumptions met
- Asking, What happens after technology shock?
- Find:
  - SVAR procedure does not uncover model's impulse response
  - Having capital in model requires infeasibly many VAR lags
  - Earlier equivalence results imply that SVARs are not useful guides

What You Get from SVAR Procedure

- Structural MA

$$X_t = A_0\epsilon_t + A_1\epsilon_{t-1} + A_2\epsilon_{t-2} + \ldots, \; E\epsilon_t\epsilon_t' = \Sigma$$

What You Get from SVAR Procedure

- Structural MA for ϵ=['technology shock', 'demand shock']$'$

$$X_t = A_0\epsilon_t + A_1\epsilon_{t-1} + A_2\epsilon_{t-2} + \ldots, \; E\epsilon_t\epsilon_t' = \Sigma$$

where X_t = [Δ Log labor productivity, $(1-\alpha L)$Log hours]$'$

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where X_t = [Δ Log labor productivity, $(1-\alpha L)$Log hours]$'$

- Identifying assumptions:
  - technology and demand shocks uncorrelated $(\Sigma = I)$
  - demand shock has no long-run effect on productivity

Impulse Responses and Long-Run Restriction

- Impulse response from structural MA:

  Blip ϵ_1^d for response of productivity to demand

$$\log(y_1/l_1) - \log(y_0/l_0) = A_0(1,2)$$

$$\log(y_2/l_2) - \log(y_0/l_0) = A_0(1,2) + A_1(1,2)$$

$$\vdots$$

$$\log(y_t/l_t) - \log(y_0/l_0) = A_0(1,2) + A_1(1,2) + \ldots + A_t(1,2)$$

- Long-run restriction:

  Demand shock has no long run effect on level of productivity

$$\sum_{j=0}^{\infty} A_j(1,2) = 0$$

Deriving Structural MA from VAR

- OLS regressions on bivariate VAR: $B(L)X_t = v_t$

$$X_t = B_1 X_{t-1} + B_2 X_{t-2} + B_3 X_{t-3} + B_4 X_{t-4} + v_t, \quad E v_t v_t' = \Omega$$

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- Invert to get MA: $X_t = B(L)^{-1} v_t = C(L) v_t$

$$X_t = v_t + C_1 v_{t-1} + C_2 v_{t-2} + \ldots$$

with $C_j = B_1 C_{j-1} + B_2 C_{j-2} + \ldots + B_j, \quad j = 1, 2, \ldots$

Identifying Assumptions

- Work from $X_t = v_t + C_1 v_{t-1} + C_2 v_{t-2} + \ldots,\ E v_t v_t' = \Omega$

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- Identifying assumptions determine 7 parameters in A_0, Σ
  - Structural shocks ϵ are orthogonal, $\Sigma = I$
  - Demand shocks have no long-run effect on labor productivity

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- Identifying assumptions determine 7 parameters in A_0, Σ
  - Structural shocks ϵ are orthogonal, $\Sigma = I$
  - Demand shocks have no long-run effect on labor productivity

  $\Rightarrow$ 7 equations ($A_0 \Sigma A_0' = \Omega$, $\Sigma = I$, $\sum_j A_j(1,2) = 0$)

Use Growth Model Satisfying 3 Key SVAR Assumptions

- Only 2 shocks ("technology," "demand")
- Shocks are orthogonal
- Technology shock has unit root, demand shock does not

This is the best case scenario for SVARs

Specification of Shocks in the Model

- Technology shock is efficiency wedge $A = z^{1-\theta}$

$$\log z_t = \mu_z + \log z_{t-1} + \eta_{zt}$$

- "Demand" shock is labor wedge

$$\tau_{lt} = (1-\rho)\bar{\tau}_l + \rho\tau_{lt-1} + \eta_{\tau t}$$

- With 3 key SVAR assumptions imposed
  - Only 2 shocks ("technology," "demand")
  - Shocks orthogonal ($\eta_z \perp \eta_\tau$)
  - Technology shock has unit root, demand shock does not

Our Evaluation of SVAR Procedure

- Use growth model satisfying SVAR's 3 key assumptions
- Model has **theoretical** impulse response

$$X_t = D(L)\eta_t$$

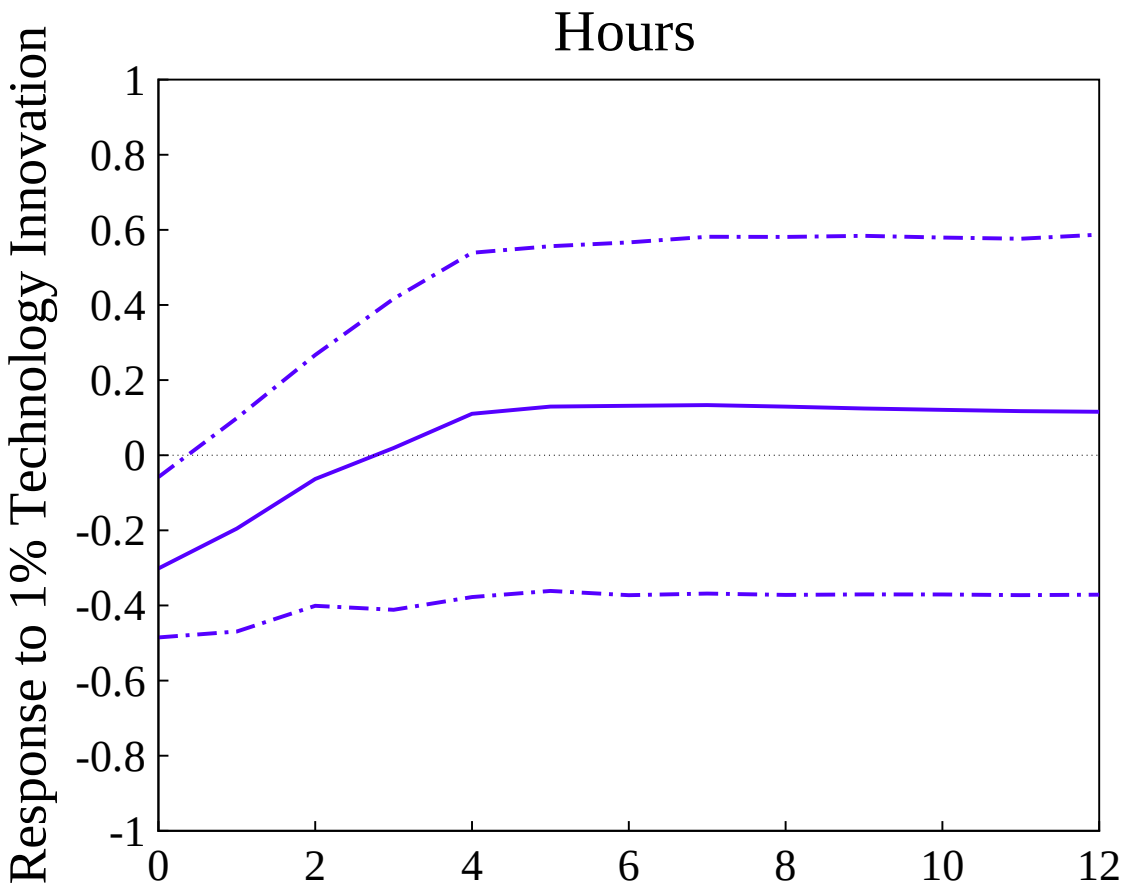
- Generate data from model
- Apply SVAR to this data to get **empirical** impulse response

$$X_t = A(L)\epsilon_t$$

- Compare model impulse responses with SVAR responses

Application to Post-war Business Cycles

DSVAR($\alpha = 1$) Procedure Applied to US Data

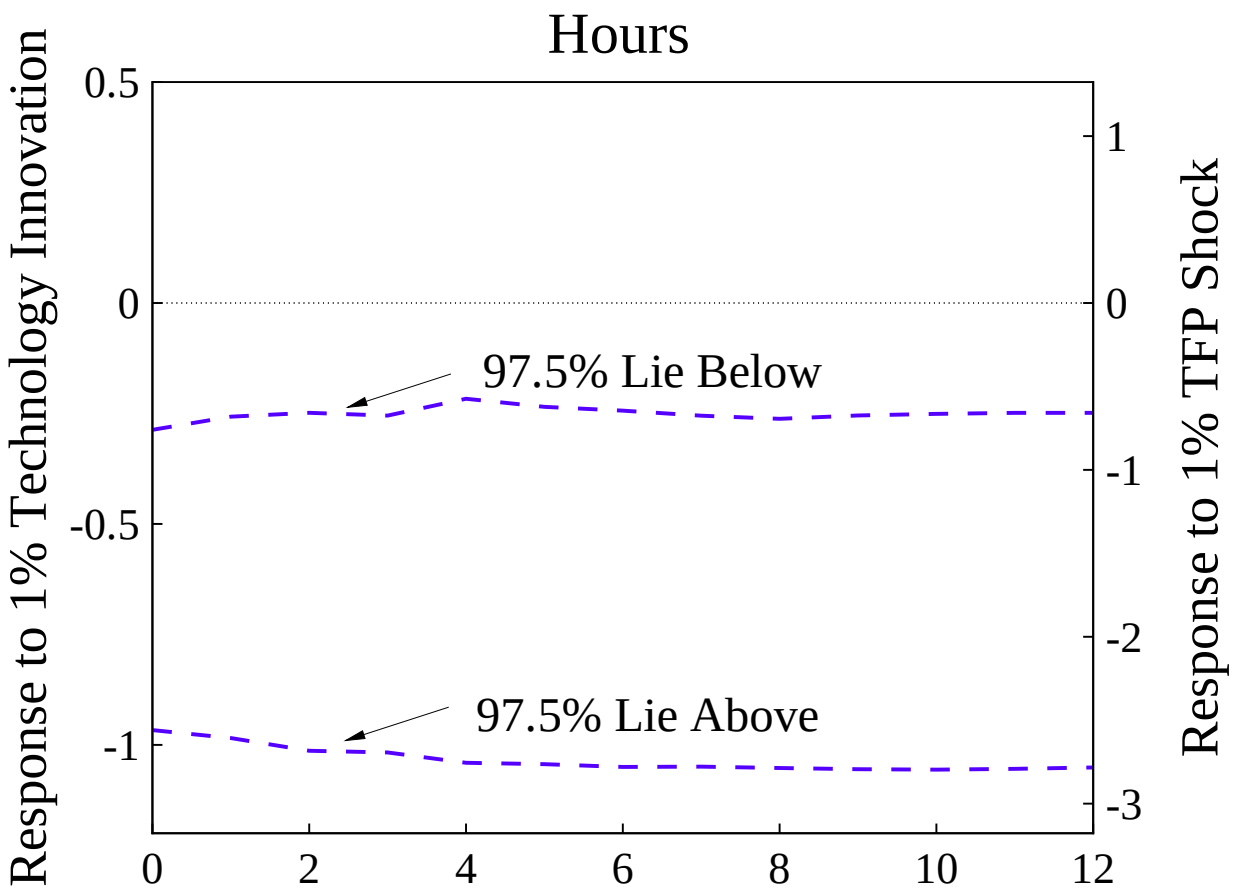


Hours fall after a positive shock in technology

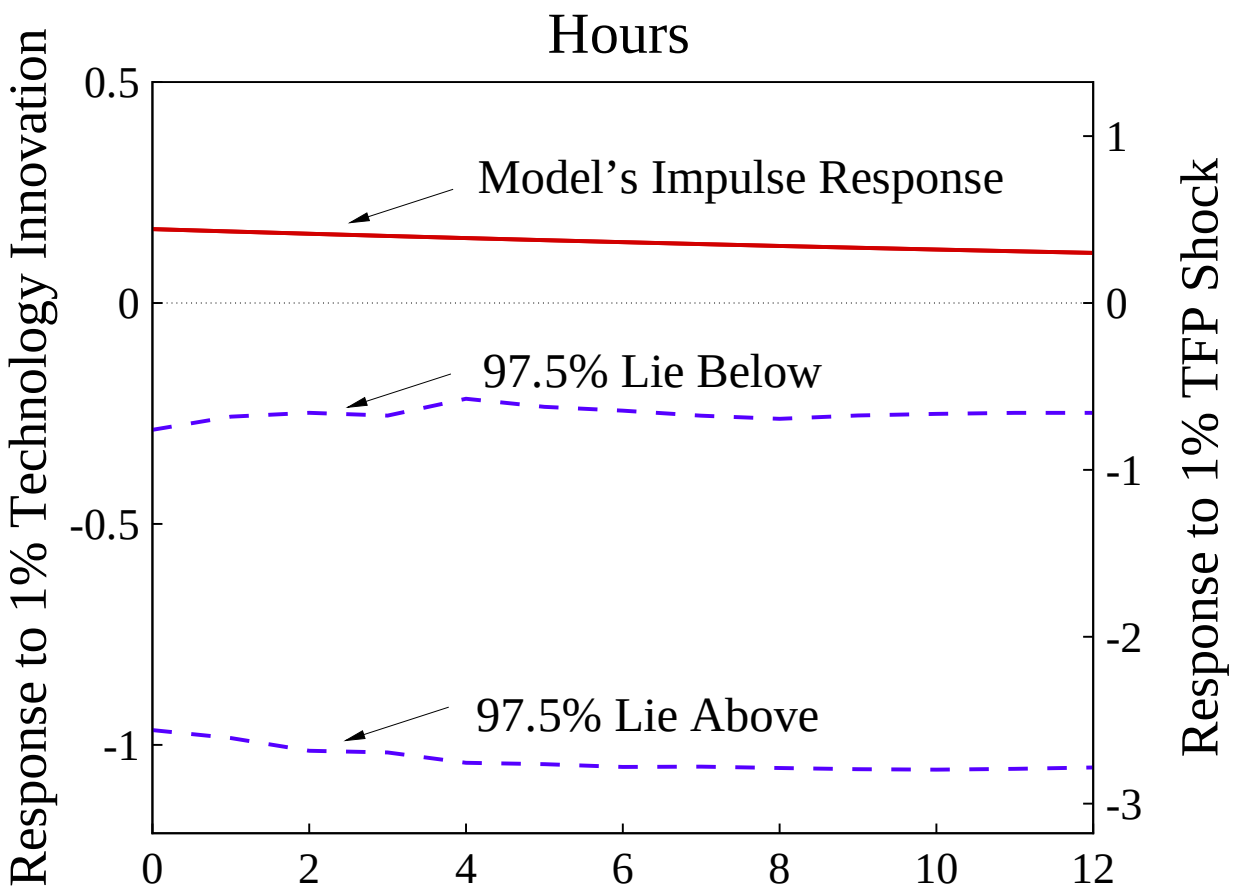
Apply SVAR Procedure to Data from Model

- Draw many sequences of η of length 180
- Use model to derive many sequences for labor productivity and hours
- Run statistical tests (e.g., unit root, lag-length specification)
- Apply SVAR procedure

DSVAR Impulse Responses using Model Generated Data



Compare DSVAR Responses and Model Responses



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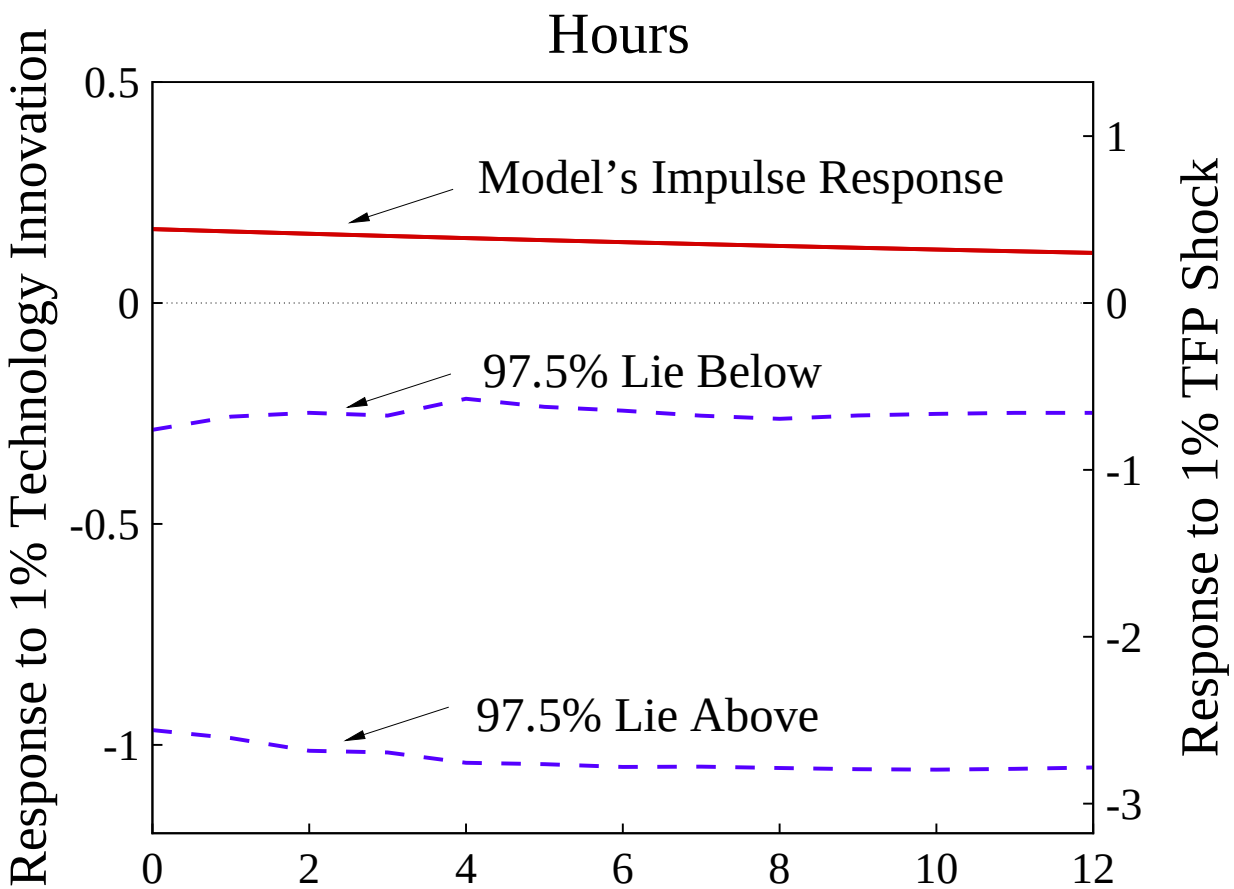


SVAR leads to mistaken inference

Recap

- Used model satisfying SVAR 3 key assumptions
- Compared DSVAR responses and model responses
- SVAR procedure leads to mistaken inference

Recap

- Used model satisfying SVAR 3 key assumptions
- Compared DSVAR responses and model responses
- SVAR procedure leads to mistaken inference
- Now explain why

Analyze Mistaken Inference

- Implicit assumptions of SVAR
  - MA representation is invertible
  - Few AR lags enough (e.g., 4)
- Violation of either leads to mistaken inference
  - Easy to get around invertibility problem from over-differencing
  - Hard to get around invertibility problems a la Hansen-Sargent
  - Hard to get around short lag length problem

Invertibility due to Overdifferencing

- Well-known that over-differencing leads to noninvertibility

- Trivial example: $\log l_t = \eta_t$, η i.i.d

$$(1 - L) \log l_t = (1 - L)\eta_t$$

is not invertible

- Easy fix is quasi-differencing:

$$(1 - \alpha L) \log l_t = (1 - \alpha L)\eta_t$$

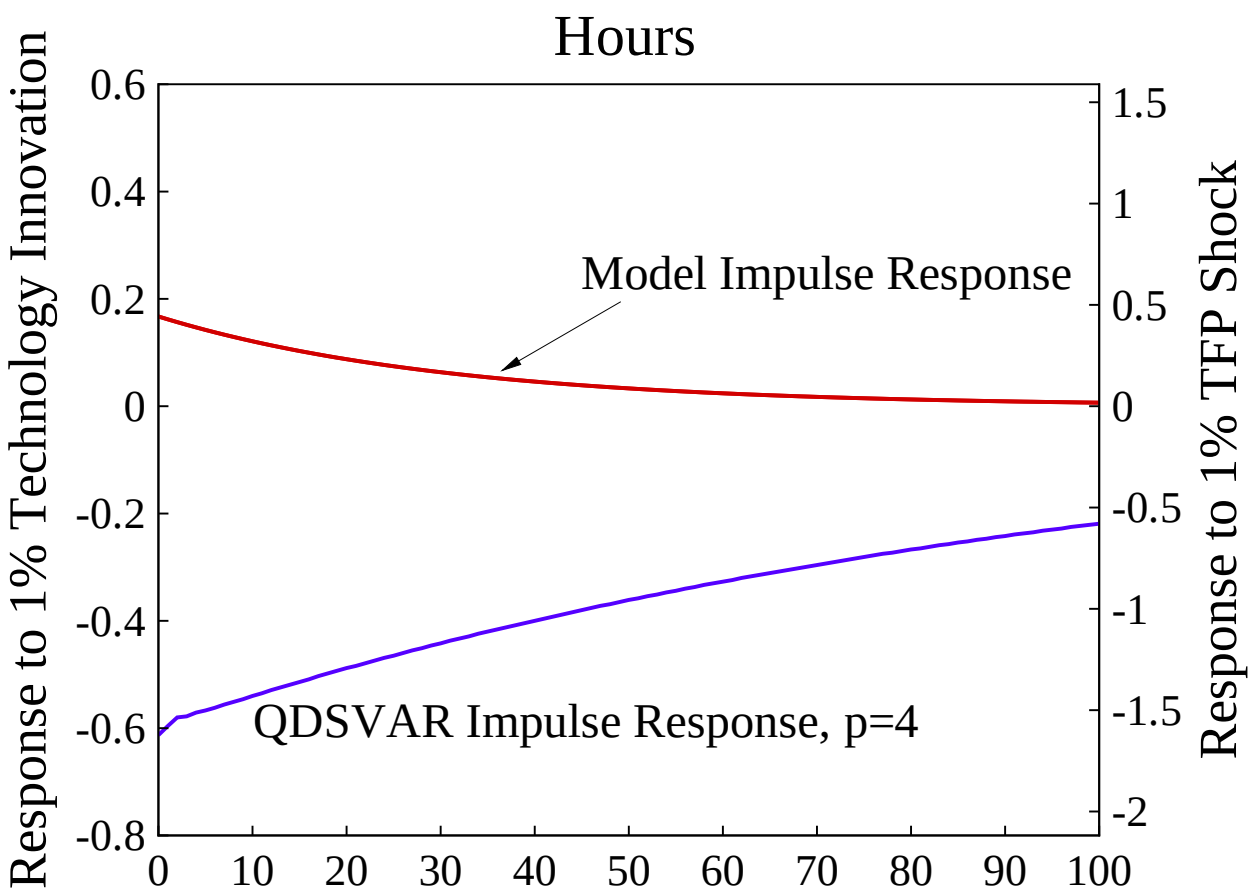
is invertible for any $\alpha < 1$

The Short Lag Length Problem

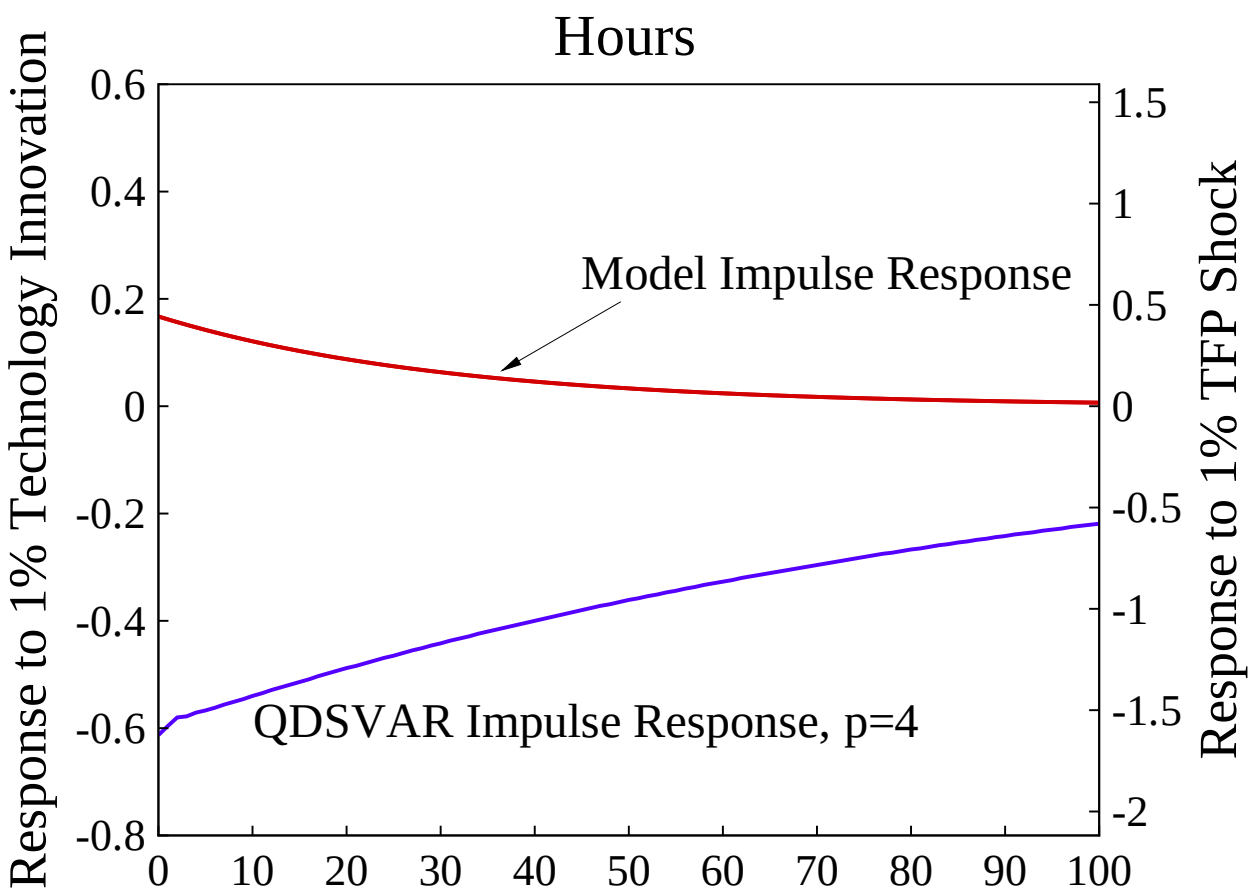
- Use
  - Quasi-differencing (QDSVAR) to avoid invertibility problem
  - 100,000 sample to avoid small sample problem

and see if we can uncover model's impulse response

QDSVAR($\alpha = .99$) with 100,000 Sample

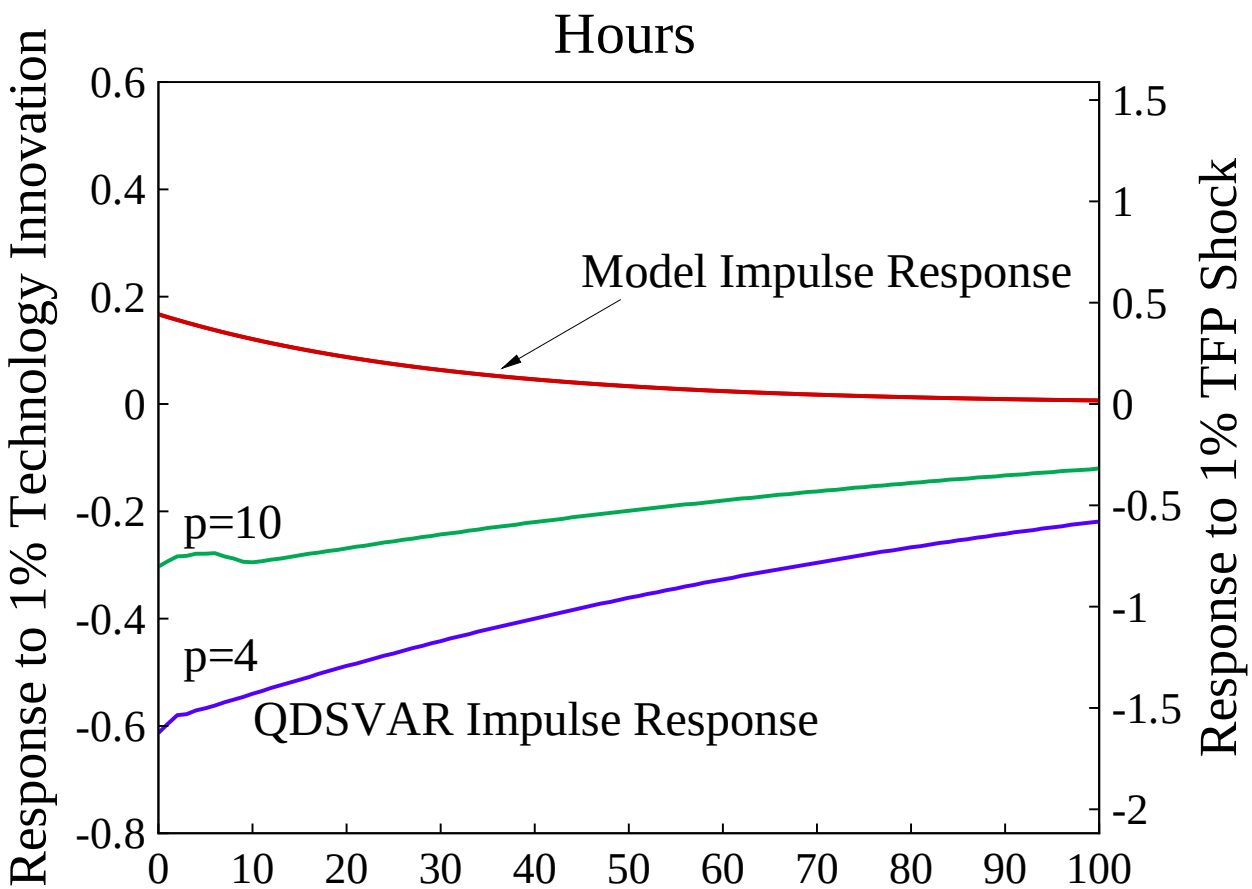


QDSVAR($\alpha = .99$) with 100,000 Sample



QDSVAR doesn't uncover model's impulse response

Short Lag Length Leads to Mistaken Inference



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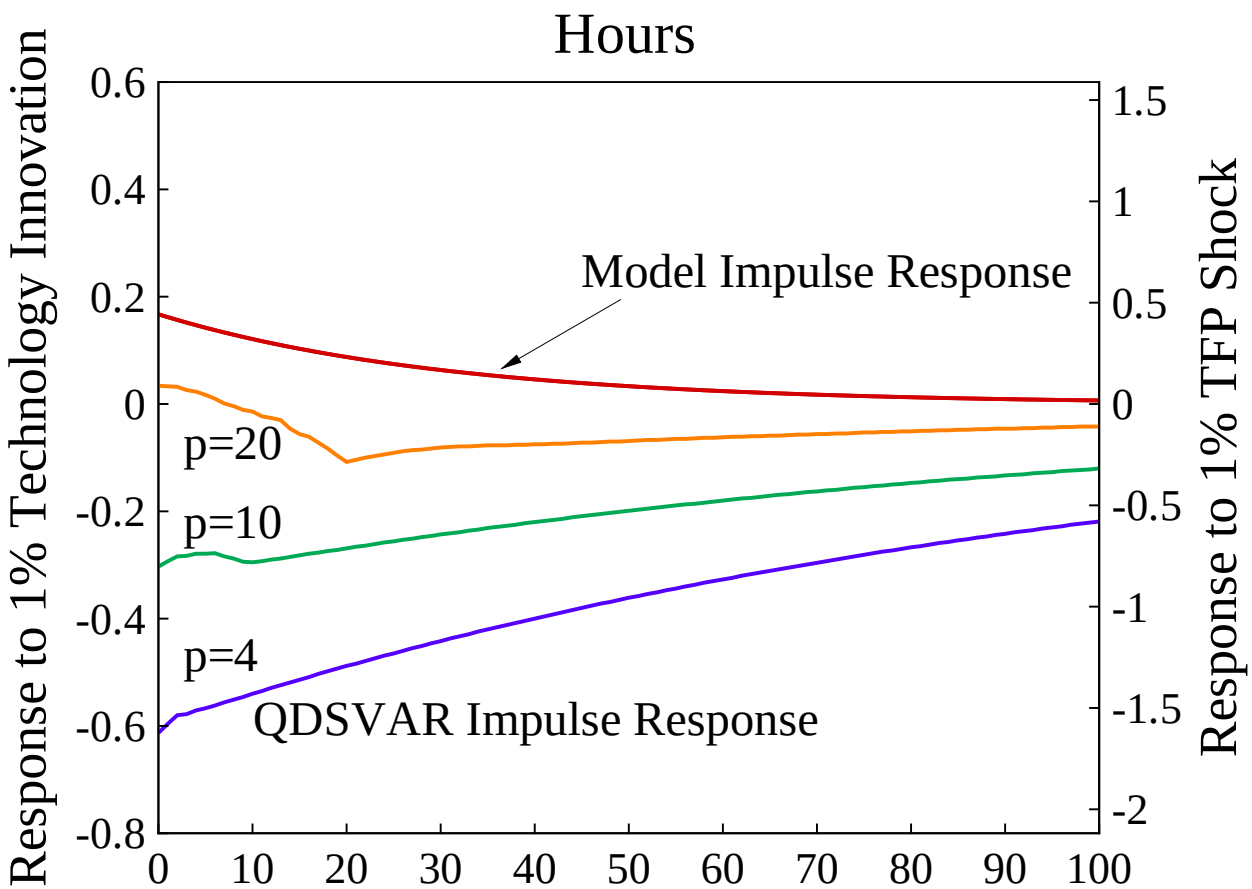

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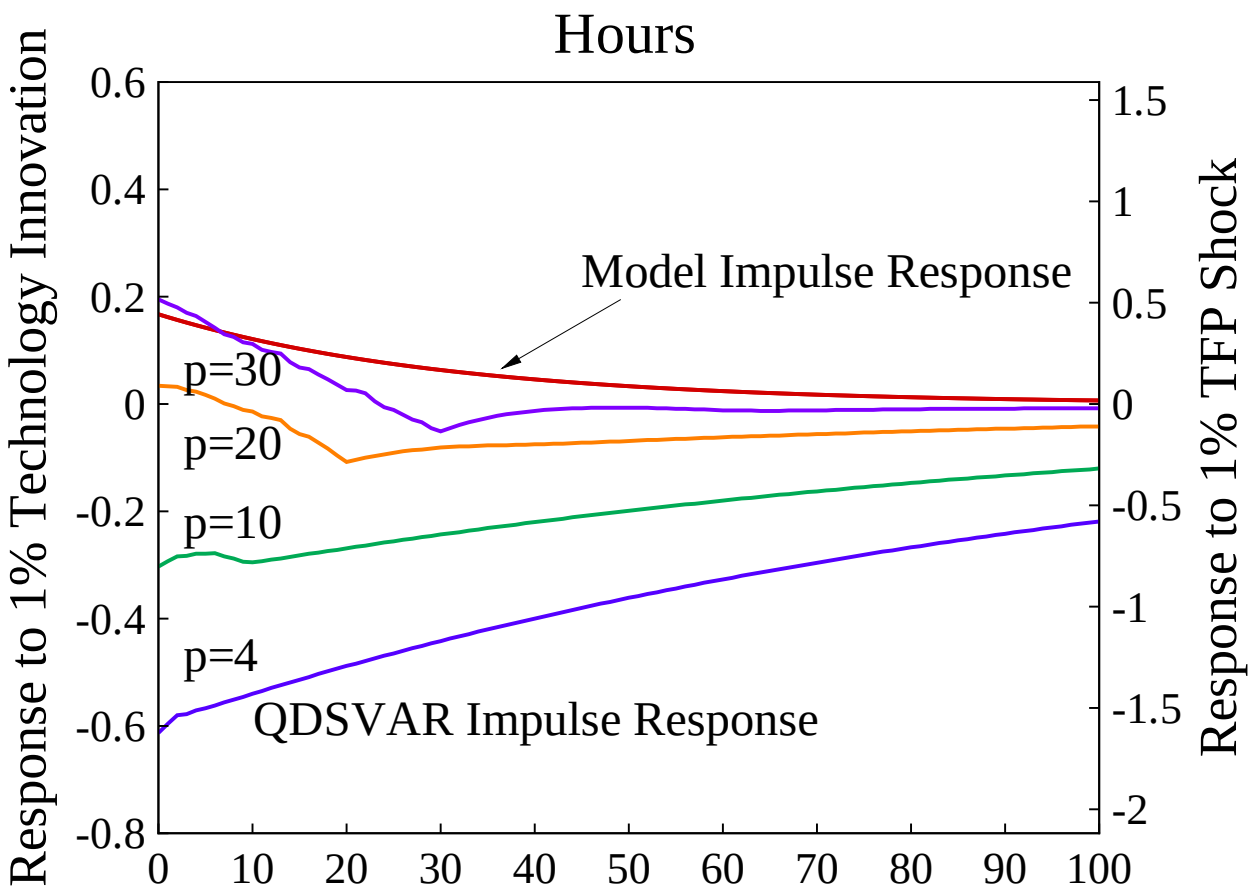

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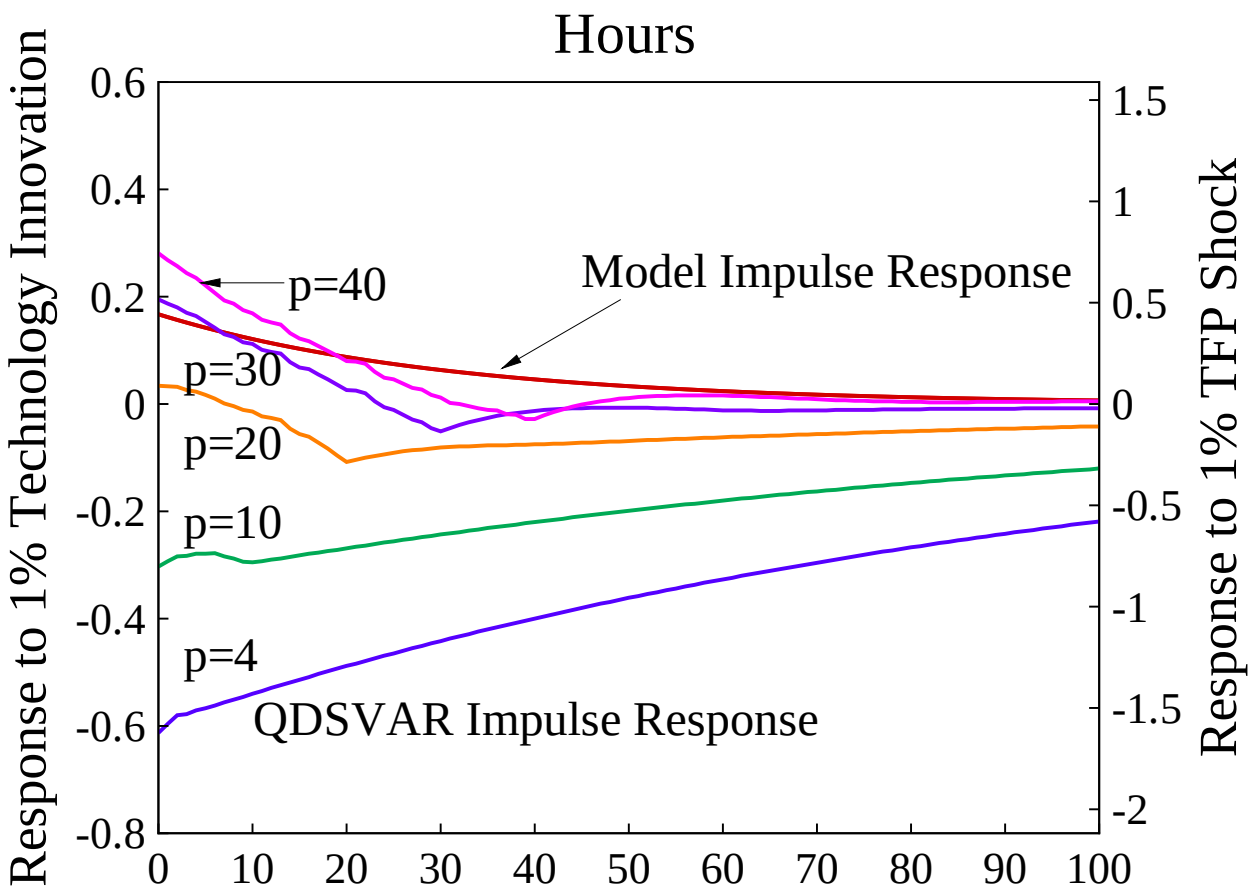

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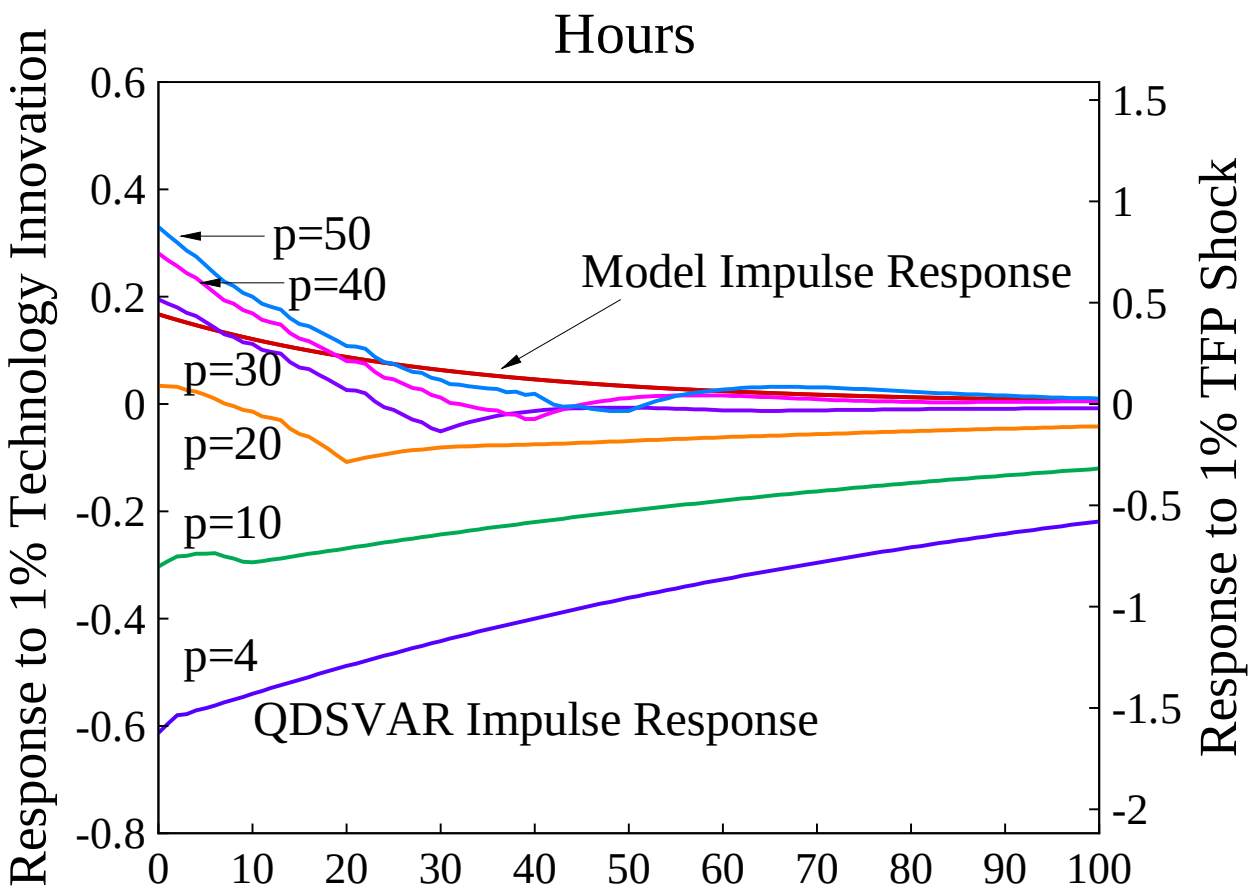

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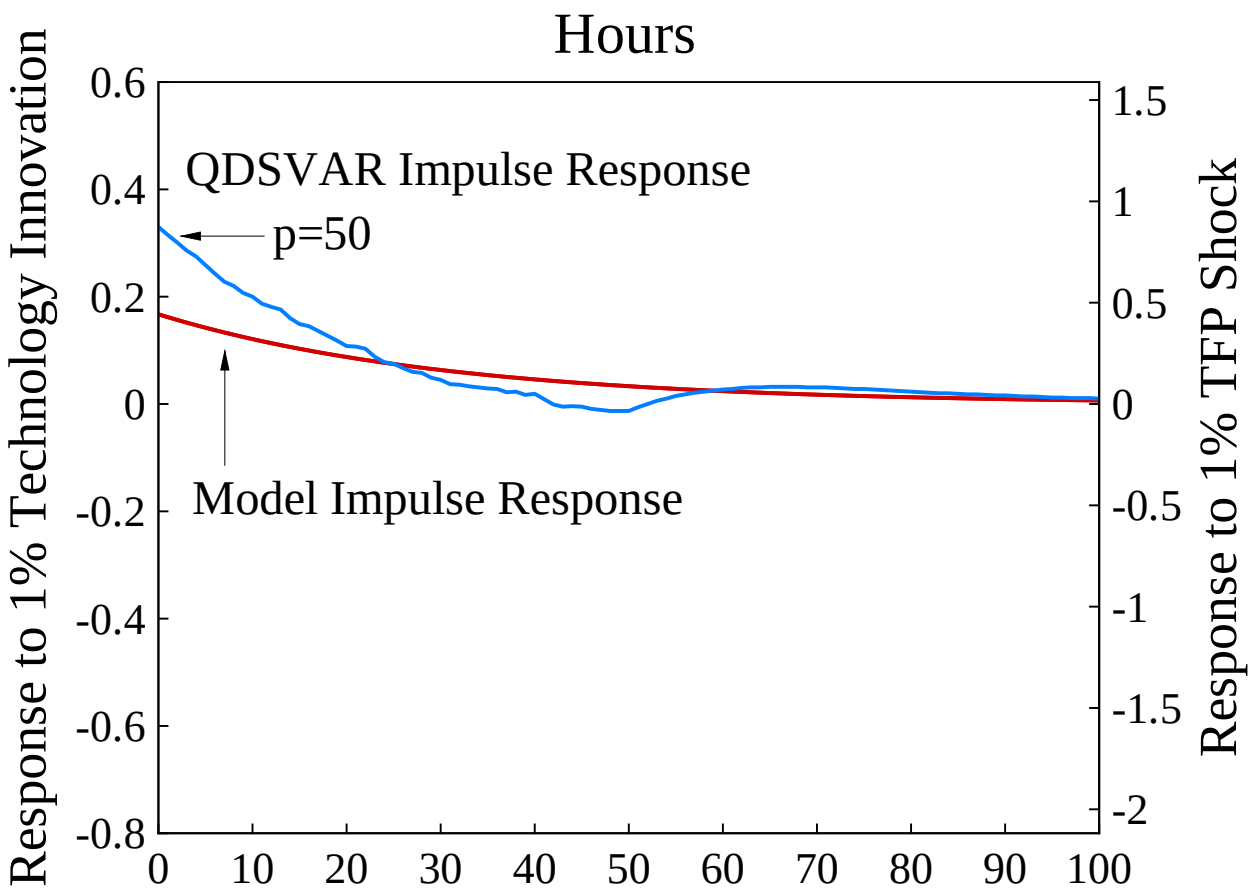
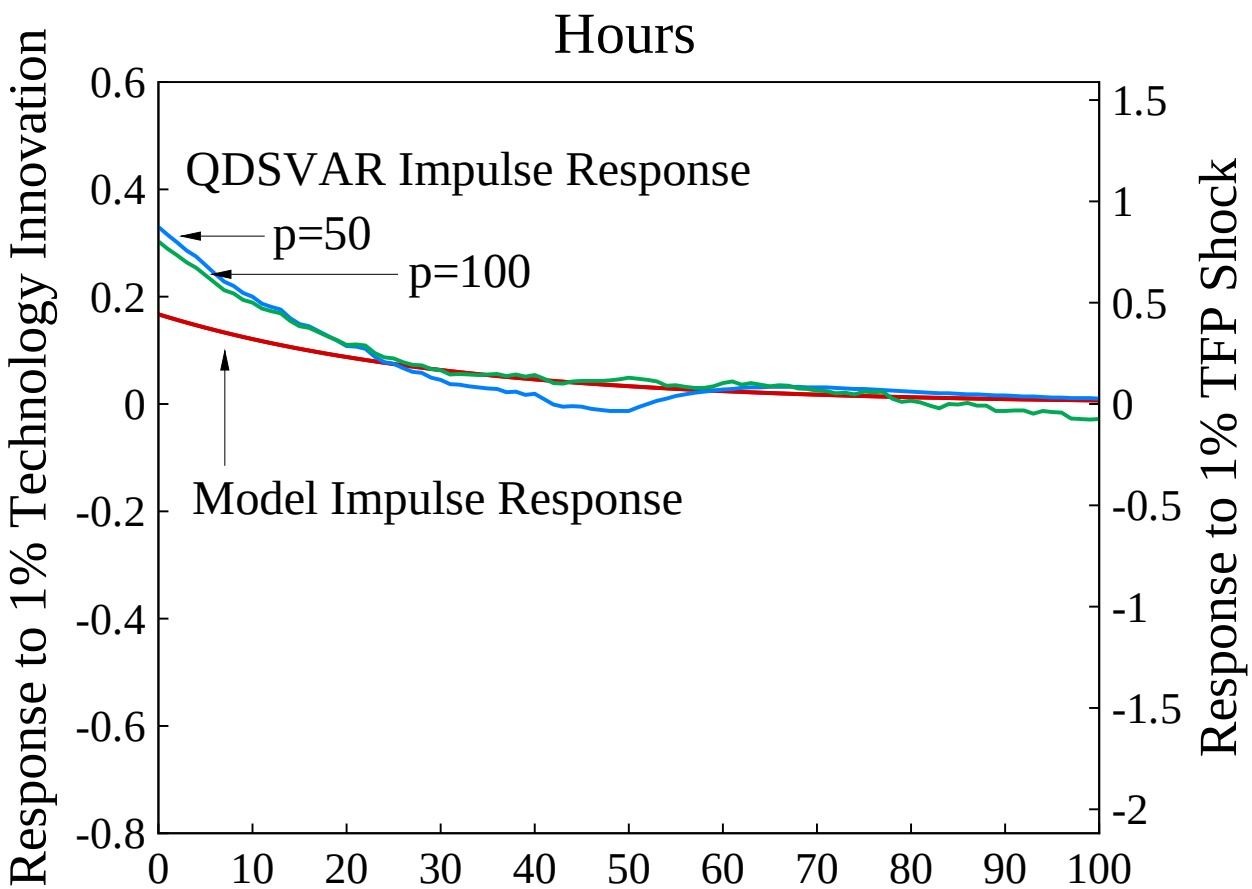


Overshoots and then comes back.....

Short Lag Length Leads to Mistaken Inference



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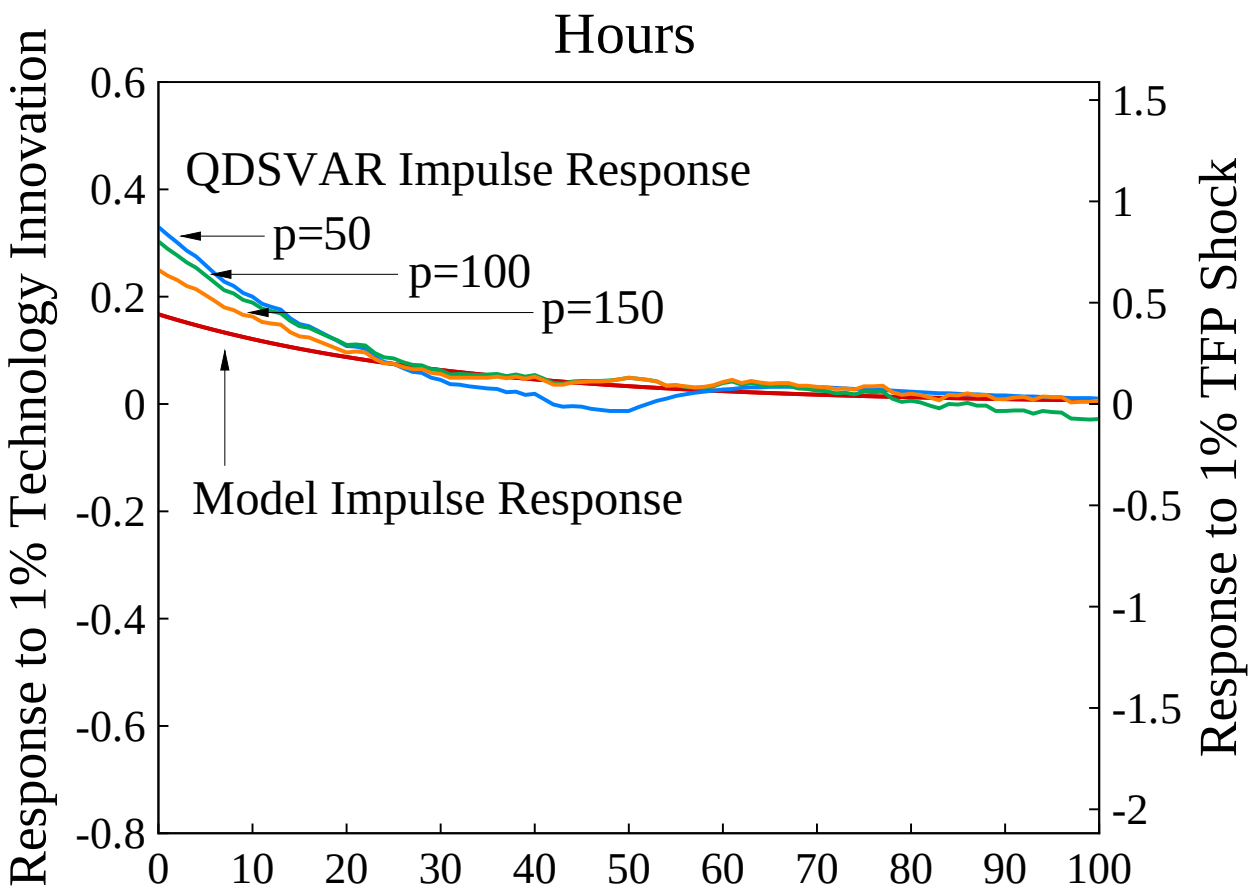

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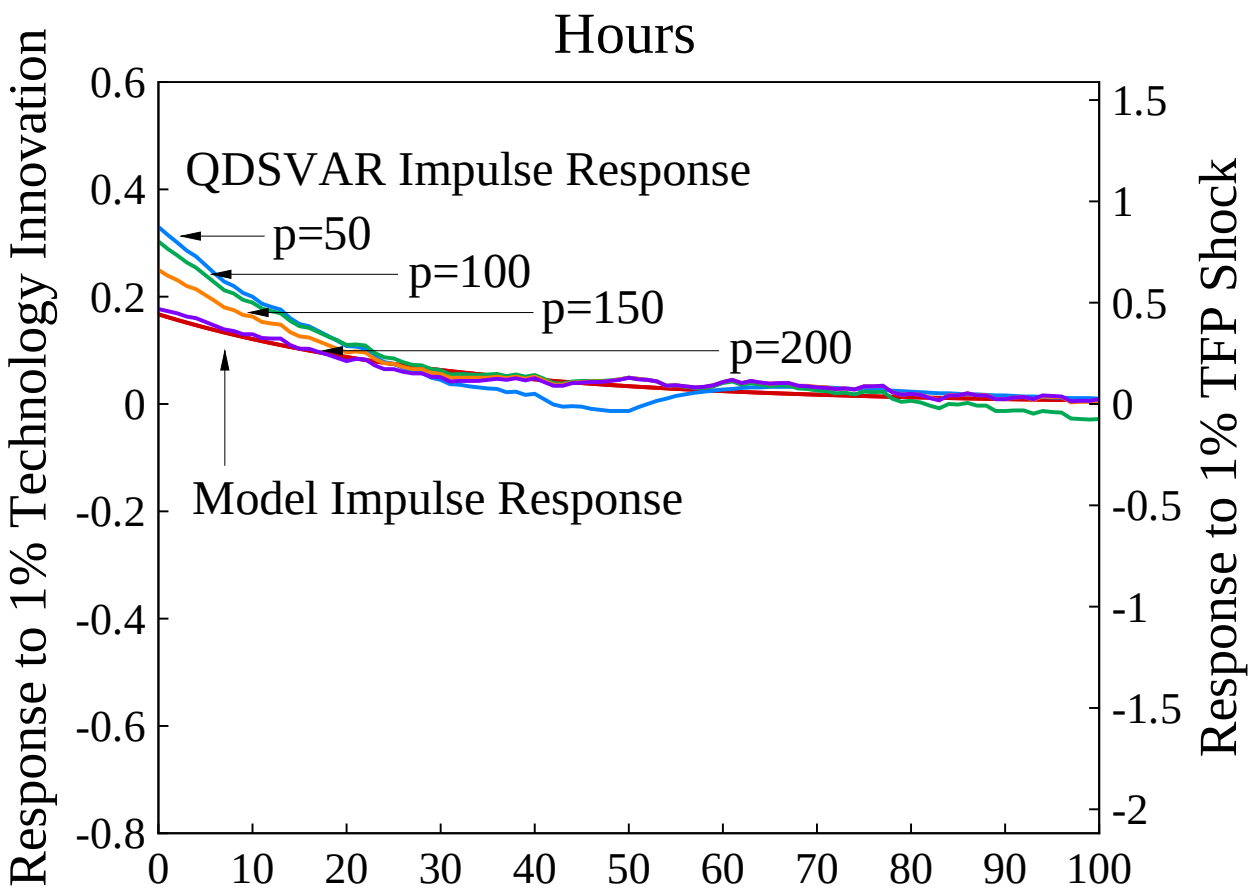


Long AR Needed Because of Capital

- Capital decision rule, with $\hat{k}_t = k_t / z_{t-1}$:

$$\log \hat{k}_{t+1} = \gamma_k \log \hat{k}_t + \gamma_z \eta_{zt} + \gamma_\tau \tau_t$$

- So others, like l_t, have ARMA representation

$$\log l_t = \gamma_k \log l_{t-1} + \phi_z (1 - \kappa_z L) \eta_{zt} + \phi_\tau (1 - \kappa_\tau L) \tau_t$$

- What does the AR representation, $B(L) X_t = v_t$, look like?

Model Has Infinite-order AR

- Proposition: Model has VAR coefficients B_j such that

$$B_j = MB_{j-1}, \quad j \geq 2,$$

where M has eigenvalues equal to α (the differencing parameter) and

$$\left(\frac{\gamma_k - \gamma_l \phi_k / \phi_l - \theta}{1 - \theta} \right)$$

γ_k, γ_l are coefficients in the capital decision rule

ϕ_k, ϕ_l are coefficients in the labor decision rule

- Eigenvalues of M are α and .97 for the baseline parameters

What Happens with Two Few Lags

- From SVAR procedure, want to recover model's:
  - Variance-covariance matrix Ω_m
  - Sum of MA coefficients $\bar{C}_m$
- Example: Run VAR with 1 Lag and see what SVAR recovers
  - Variance-covariance matrix (with $V(X) = EXX'$):

$$\Omega = \Omega_m + M\left(\Omega_m - \Omega_m V(X)^{-1}\Omega_m\right)M'$$

  - (Inverse of) sum of MA coefficients:

$$\bar{C}^{-1} = \bar{C}_m^{-1} + M(I-M)^{-1}C_{m,1} + M\left(\Omega_m - V(X)\right)V(X)^{-1}$$

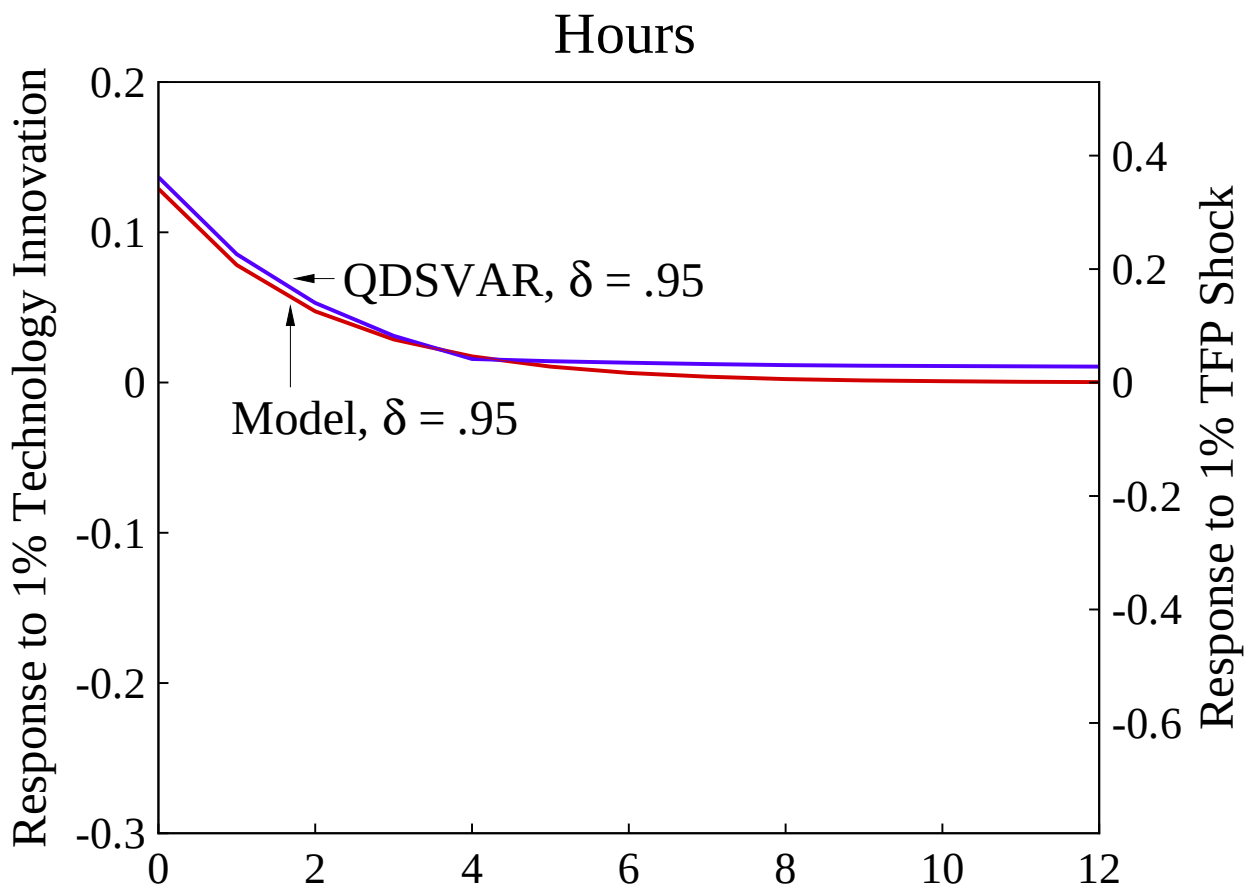
Notice that M is important factor in garbled terms!

Proposition for Two Special Cases

- Proposition: If model has
  - no capital ($\theta = 0$) or
  - only one shock ($\sigma_\tau = 0$),

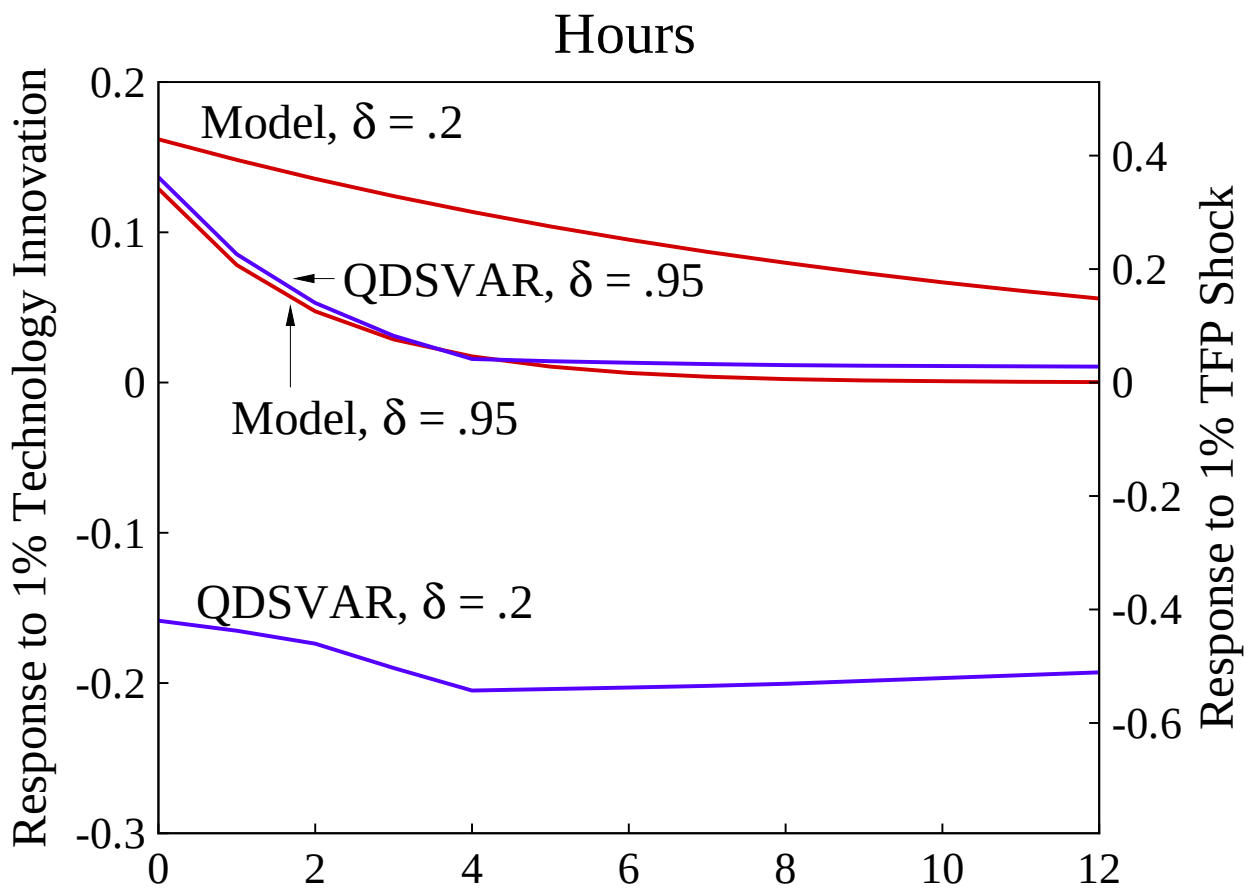
  then there is no mistaken inference.

What if Capital Less Important?



4 lags do fine if depreciation very high

What if Capital Less Important?



4 lags do fine if depreciation very high

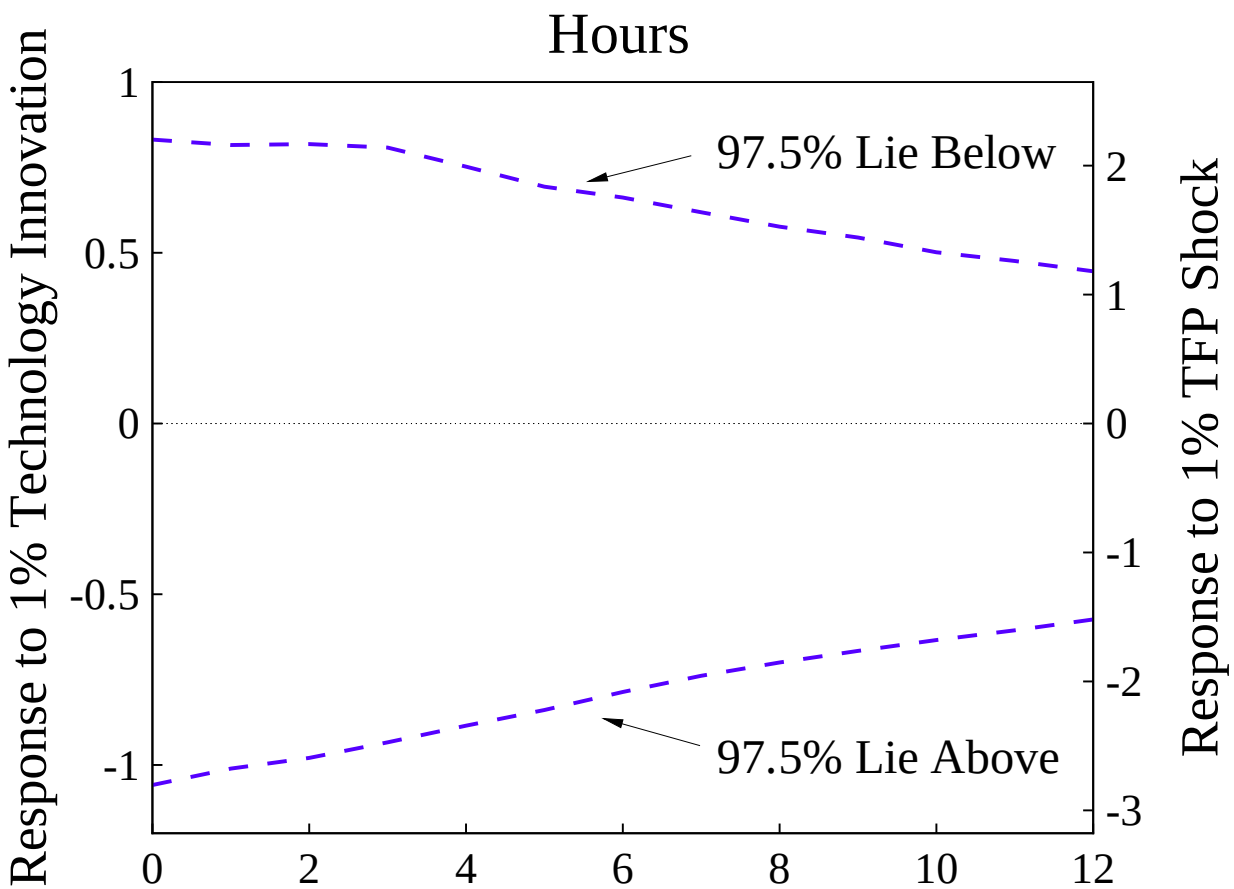
Recap

- Used model satisfying 3 key assumptions
- SVAR procedure leads to mistaken inference
- Short lag length is the problem

Recap

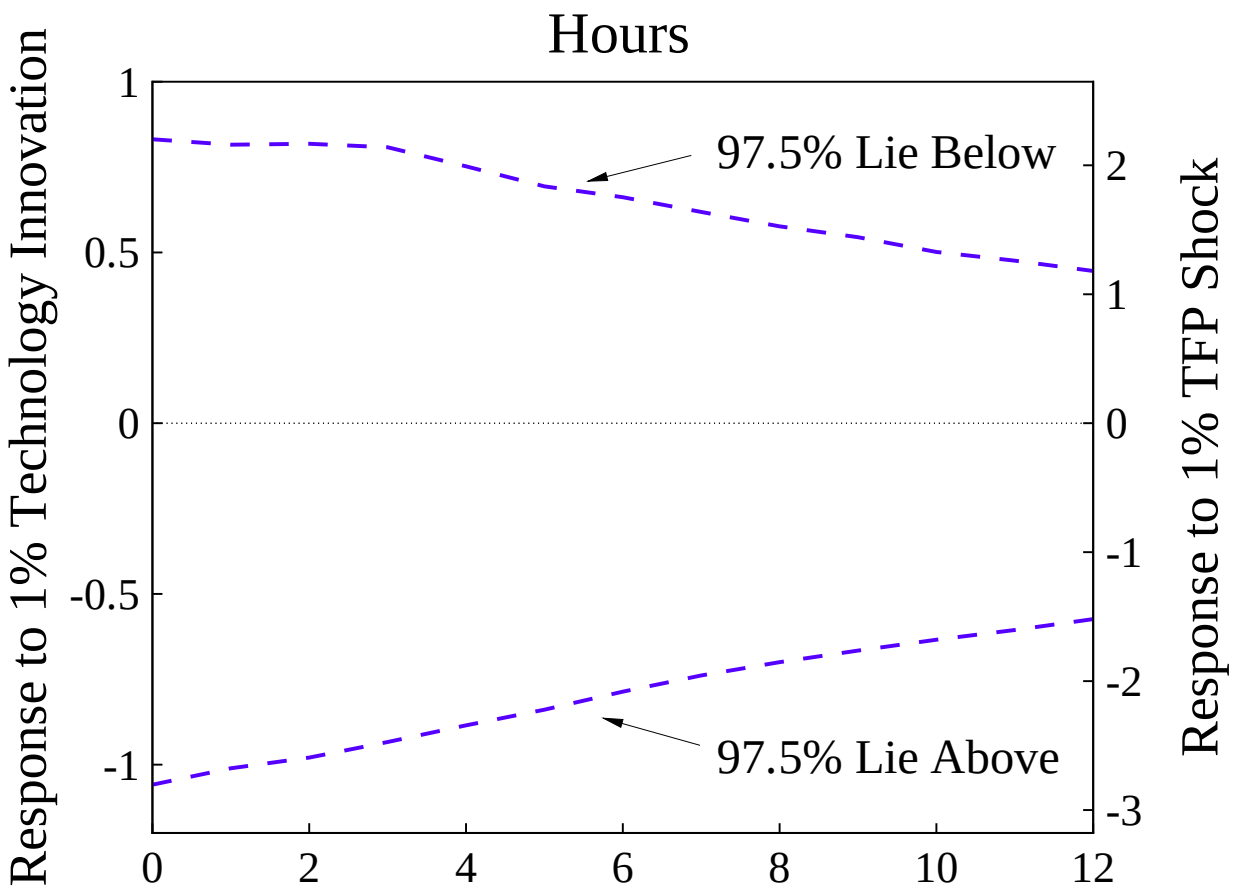
- Used model satisfying 3 key assumptions
- SVAR procedure leads to mistaken inference
- Short lag length is the problem
- What happens if we evaluate SVAR with hours in levels (LSVAR)?

Short Sample Result: LSVAR Uninformative



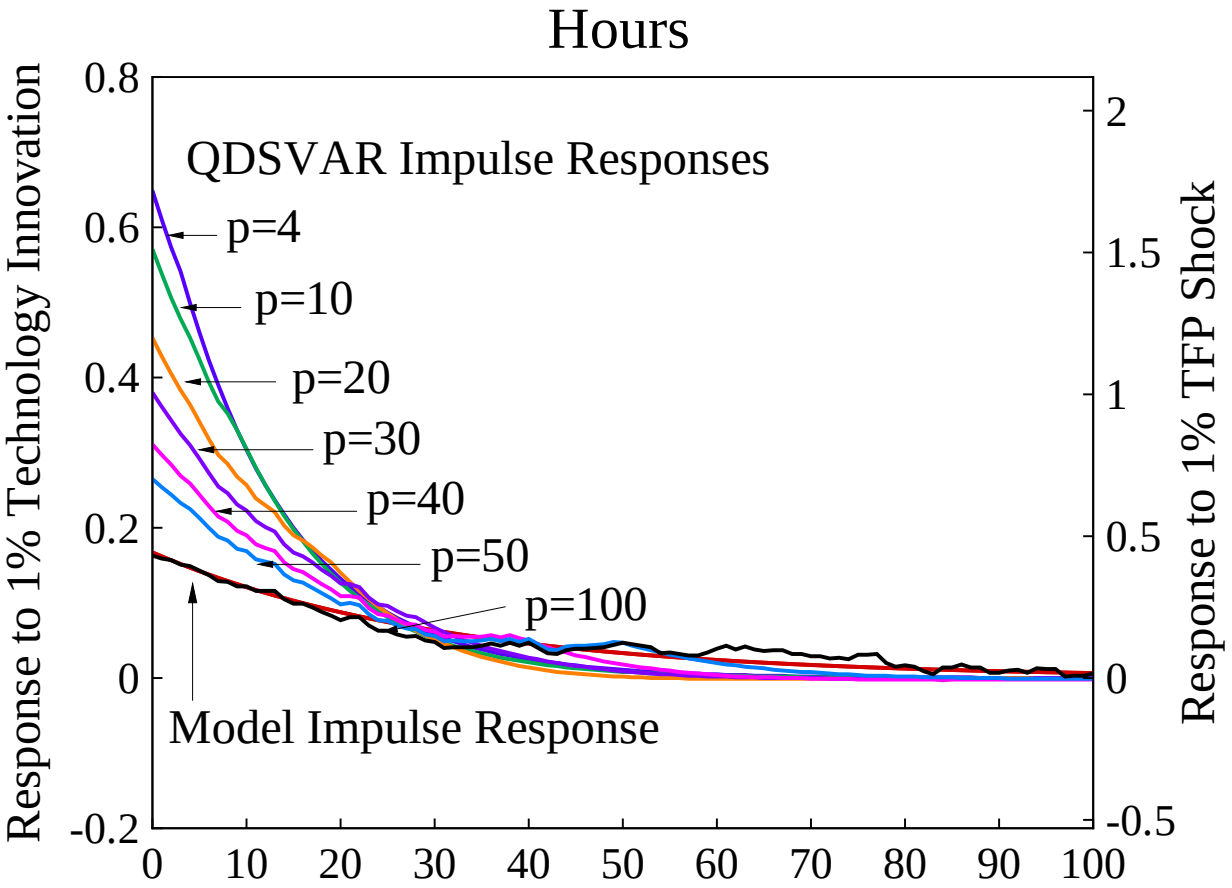
1000 applications of SVAR on model data yields huge range

Short Sample Result: LSVAR Uninformative



Confidence band of HP-filtered output due to technology: [7%,97%]

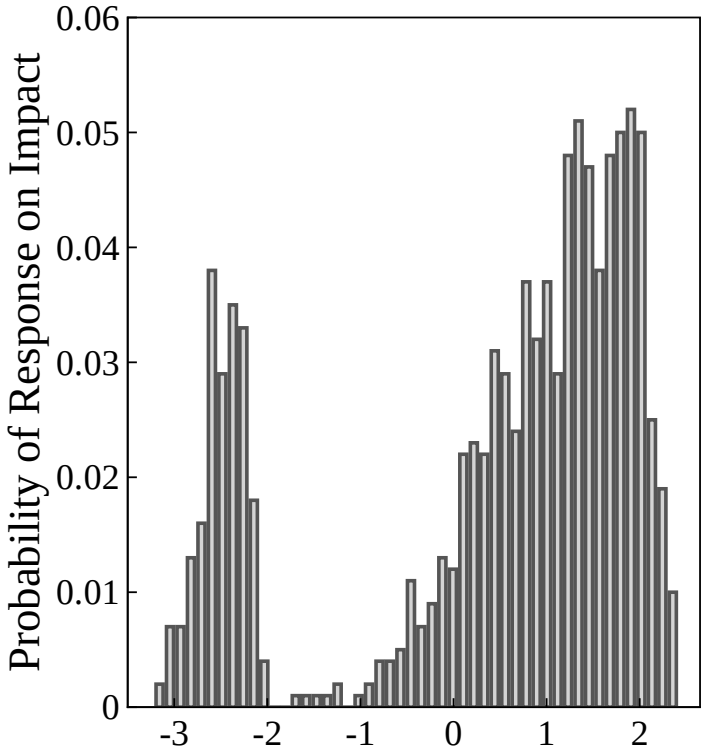
Long Sample Results: Long Lags Required

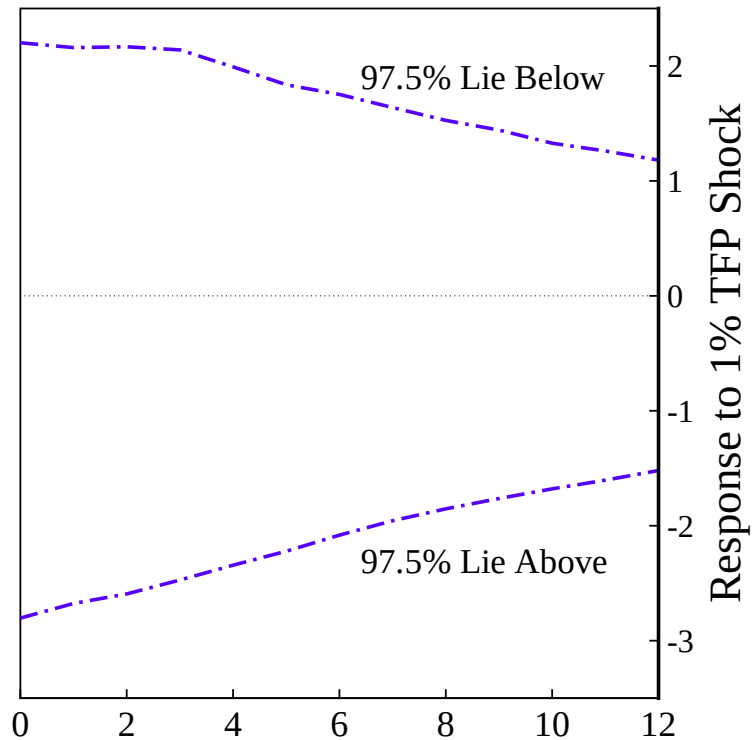


Propositions for LSVARs

- Same as before:
  - Model has infinite-order VAR
  - No mistaken inference if there is no capital or 1 shock
- One other result is critical for understanding short sample results

No Robust Prediction for Sign on Impact





Proposition: nonzero SVAR error $\Rightarrow$ impact sign changes as vary σ_τ

Recap

- Short lag length leads to mistaken inference in SVARs

- Problem not fixed easily given available data

Recap

- Short lag length leads to mistaken inference in SVARs
- Problem not fixed easily given available data
- What does theory point to?

What Does Theory Point To?

- Business cycle models have state space representations:

$$\log \hat{k}_{t+1} = \gamma_k \log \hat{k}_t + \gamma_s' s_t$$

$$s_{t+1} = P s_t + Q \eta_{t+1}$$

with shocks s_t (e.g., $\log z_t$, τ_t). Write as VAR:

$$S_{t+1} = F S_t + G \eta_{t+1}$$

- with the measurement equation

$$X_t = H S_t + \omega_t$$

X_t are observations, S_t is the state, and ω_t is measurement error

- Why not estimate this system as opposed to various VARs?

What have we learned?

- BC accounting is promising alternative to SVARs.
- Statistical methods must be guided by theory
- Empirical "facts" may indeed be fictions